

Office Hour One: Groups

An annotated study edition

Original chapter: Matt Clay and Dan Margalit, *Office Hours with a Geometric Group Theorist* (Princeton University Press, 2017), pages 3–20.

How to read this

Each spread shows one original page of the chapter on the left, untouched, and a column of annotations on the right. Orange numbered markers 1 2 3 on the page point to the line the annotation is about; a faint yellow bar marks the exact statement when it matters. The annotations aim at a strong high-school reader: they fill in background the authors assume (functions, modular arithmetic, matrices), draw pictures of every construction, work examples in full, and give hints for all thirteen exercises.

Colour code for the boxes

Big picture	where an idea sits in the whole book
Why?	the reason a definition or claim is true
Watch out	a common misreading, or a slip in the printed text
Try it	something to check by hand, and hints for exercises
Looking ahead	connections to later office hours
Background	facts from school mathematics the text assumes

Contents of the chapter

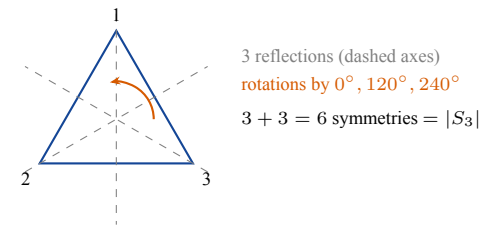
Symmetry, the cube, and S_4	book pp. 3–5	spreads 1–3, A2–A5
1.1 Groups: D_n , S_n , $\mathbb{Z}/n\mathbb{Z}$	book pp. 5–9	spreads 3–7, B, B2
1.2 Infinite groups: $\mathbb{Z}$, $\mathbb{Z}^2$, matrices, free groups	book pp. 9–13	spreads 7–11
1.3 Homomorphisms and normal subgroups	book pp. 13–17	spreads 11–15
1.4 Group presentations	book pp. 17–20	spreads 15–18
Visual pages V1–V5	—	between spreads
Chapter map, exercise hints, glossary	—	closing pages

Big picture

The one sentence to keep in mind. The whole chapter is an argument that “group” and “the symmetries of something” are the same idea seen from two sides. Every example is introduced twice: once as a set with a multiplication, and once as the symmetries of a picture you can draw. Office Hour 2 turns this into a theorem (every group is the symmetry group of its Cayley graph).

Background you need

You will get the most out of the chapter if these words already mean something to you: *function*, *bijection* (one-to-one and onto), *composition* $g \circ h$ (do h first), *remainder* on division by n , *vector addition*, 2×2 *determinant* and the fact that $\det(MN) = \det M \det N$. Each is reviewed in a grey box the first time the text needs it.



A warm-up the chapter skips: the equilateral triangle has 6 symmetries, and they behave exactly like the 6 permutations of its 3 corners.

Office Hour One

Groups

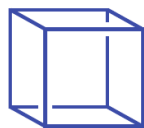
Matt Clay and Dan Margalit



Symmetry, as wide or as narrow as you define its meaning, is one idea by which man through the ages has tried to comprehend and create order, beauty and perfection.

Hermann Weyl

- 1 An important problem in mathematics is to determine the symmetries of objects. The objects we are interested in might be concrete things such as planar shapes, higher-dimensional solids, or the universe. Or they might be more abstract, such as groups, spaces of functions, or electric fields.
- 2 We'll begin in this office hour with a general discussion of groups; the focus will be on interpreting every group as a group of symmetries of some object. In the second office hour we will make precise what it means for a group to be a group of symmetries of an object. These ideas are at the very heart of geometric group theory, the study of groups, spaces, and the interactions between them.
- 3 **Symmetry.** By understanding the symmetries of an object we come closer to fully understanding the object itself. For instance, consider a cube in Euclidean 3-space:



1 “Determine the symmetries of objects”

A *symmetry* of an object is a way of moving it that leaves it looking exactly as before. The key point the whole book builds on: symmetries can be *combined* (do one, then another) and *undone*. Those two features are what the word “group” packages up.

Background you need

A *function* $f : X \rightarrow Y$ assigns to each input one output. It is a *bijection* if every element of Y is hit exactly once; then it has an inverse f^{-1} . The *composition* $g \circ h$ means “apply h , then g ”. Symmetries are bijections from an object to itself, so all of this applies to them.

2 The plan of the first two office hours

Office Hour 1 (this chapter) goes one way: start from familiar groups and show each is a set of symmetries of a picture. Office Hour 2 goes the other way and proves it in general. The phrase “the interactions between them” is the subject of geometric group theory: use the geometry of the object to learn algebra about the group, and vice versa.

Big picture

Keep a running list as you read. For each group met in this chapter, write down (a) the set, (b) the multiplication, (c) the object it is the symmetries of. By the end you should have about ten rows. That table *is* the content of the chapter.

3 The cube

The chapter’s first worked example. Before turning the page, try to answer the question yourself: how many ways can you pick up a cube, turn it, and put it back down so that it occupies the same place? (Turning is allowed; mirror-reversing is not.) Count the ones you can find; the next page shows there are 24 and, better, explains what the 24 “are”.

Try it

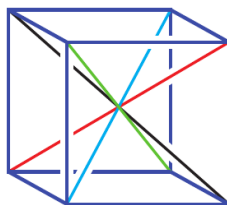
Do the same for a square drawn on paper, allowing only rotations of the paper: you should find 4. Allowing flips as well gives 8. Both answers reappear on page 6 as $\mathbb{Z}/4\mathbb{Z}$ and D_4 .

- 1 A symmetry of the cube is a rigid transformation of the cube (for now, no reflections allowed). In other words, a symmetry of the cube is what you get if you pick the cube up, rotate it with your hands, and then put it back in the same spot where it started. One way to obtain a symmetry is to skewer the cube through the centers of two opposite faces and rotate by any multiple of $\pi/2$. We could also skewer the cube through the midpoints of two opposite edges, or through two opposite corners. Here is a question:

What are all of the symmetries of the cube?

If we don't know the answer to this question, then we don't know the cube!

- 2 There is a fairly straightforward analysis which tells us that there are exactly 24 symmetries of the cube: there are eight places to send any given corner, and once a corner is fixed, there are three choices for how to turn the cube at that corner. This is a start, but leaves a lot to be desired: What is a good way to list the 24 symmetries? If we do one symmetry followed by another symmetry, we get a third symmetry; where is this third symmetry on our list?
- 3 Here is a simple idea that completely illuminates the problem. Draw the four long diagonals of the cube:



We can check the following two facts:

- 4
- Any symmetry of the cube gives a permutation of the four long diagonals.
 - Any permutation of the four long diagonals gives a unique symmetry of the cube.

The validity of the first statement is not too hard to see. The second one takes a little thought. First notice that if we take the permutation that swaps two long diagonals, then we obtain a symmetry of the cube, namely, a symmetry obtained by skewering the cube through the midpoints of the two edges that connect the endpoints of the two long diagonals in question. But an arbitrary permutation of the four long diagonals can be obtained by swapping two long diagonals at a time, so we are done.

- 6 It follows that the set of symmetries of the cube is in bijection with the set of permutations of its four long diagonals. There are $4! = 24$ permutations of the four long diagonals, agreeing with our calculation that there are 24 symmetries of the cube. Moreover, since a permutation is just a bijective function from a set to itself, we know what happens when we do one permutation, then another—we get the composition of the two functions. We'll say more about permutations in a few minutes, when we talk about symmetric groups.

1 “Rigid transformation, no reflections”

Rigid means distances are preserved: the cube is moved as a solid body, not squashed. Reflections are excluded because you cannot perform a mirror image with your hands; a transformation you *can* perform is called *orientation-preserving*. If reflections were allowed the count would double to 48.

2 The count $8 \times 3 = 24$

Fix one corner A . A symmetry must send A to one of the 8 corners. Once you know where A goes, the cube can still spin about the long diagonal through that corner, and there are exactly 3 positions (rotate by 0° , 120° , 240°) that fit. Every symmetry is determined by these two choices, so there are $8 \cdot 3 = 24$. The complaint that follows is the important one: counting is not understanding. We want to know how the 24 combine.

3 The four long diagonals

A long diagonal joins a corner to the opposite corner through the centre of the cube. See Interlude A2, two spreads on, for a labelled picture: $d_1 = AG$, $d_2 = BH$, $d_3 = CE$, $d_4 = DF$.

4 The two facts

First fact. A symmetry moves corners to corners and keeps opposite corners opposite, so it sends each long diagonal to a long diagonal; that is a rearrangement (permutation) of $\{d_1, d_2, d_3, d_4\}$.

Second fact, existence part. Take two diagonals, say $d_1 = AG$ and $d_2 = BH$. The edge AB joins an endpoint of each, and so does the opposite edge GH . Rotating by 180° about the axis through the midpoints of AB and GH swaps $A \leftrightarrow B$ and $G \leftrightarrow H$, hence swaps $d_1 \leftrightarrow d_2$, and it fixes the other two diagonals. So every “swap two diagonals” is realised; and any permutation is a product of swaps (page 7), so every permutation is realised.

Why?

Uniqueness is quietly handled by counting: the map $\{\text{symmetries}\} \rightarrow \{\text{permutations of 4 diagonals}\}$ is onto, both sets have 24 elements, so it is a bijection. A direct argument: a symmetry fixing all four diagonals must fix or flip each of them; a little geometry shows only the identity does so.

5 “Skewering through the midpoints of two edges”

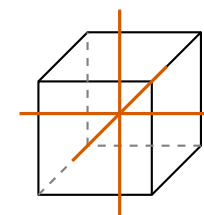
This is the edge-axis half turn drawn in Interlude A4 and listed in the dictionary of Interlude B2. There are 6 pairs of opposite edges, giving the 6 transpositions of S_4 .

6 $4! = 24$ and composition

$4! = 4 \cdot 3 \cdot 2 \cdot 1$ counts the ways to arrange 4 things in order: 4 choices for where d_1 goes, then 3, then 2, then 1. The final sentence is the real payoff: since a permutation is a function, “do one symmetry then another” becomes “compose two functions”, which is completely mechanical. The cube's symmetries are no longer a bag of 24 moves but a structured object, S_4 .

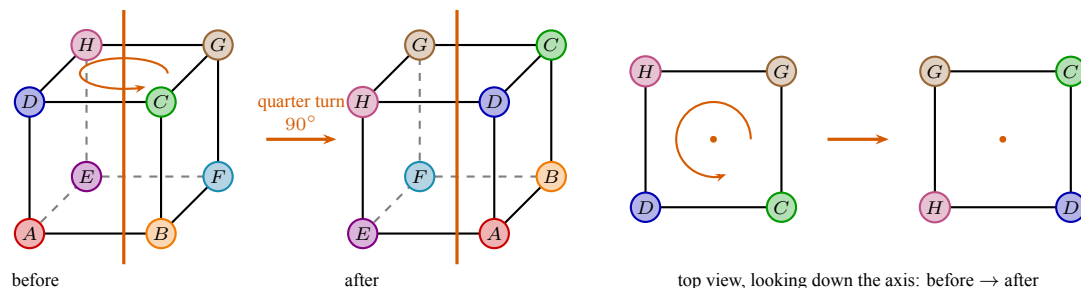
Interlude: A2. Face axes: three turns about each

Each turn is shown three ways. Left: the cube *before* (every label in its home corner) and *after* (every label drawn in the corner its point has moved to). Each corner has its own colour and the colour travels with the label, so the shifted pattern of colours shows the turn at a glance; a corner on the axis keeps its colour in place. Middle: the same turn seen from directly above the axis, where it is a flat rotation of an outline. Right: the corner moves and the resulting permutation of the diagonals $d_1 = AG$, $d_2 = BH$, $d_3 = CE$, $d_4 = DF$. The orange line is the axis, the orange arc the direction of the turn. The cube on the right shows all three face axes; the rows below use the top-bottom one.

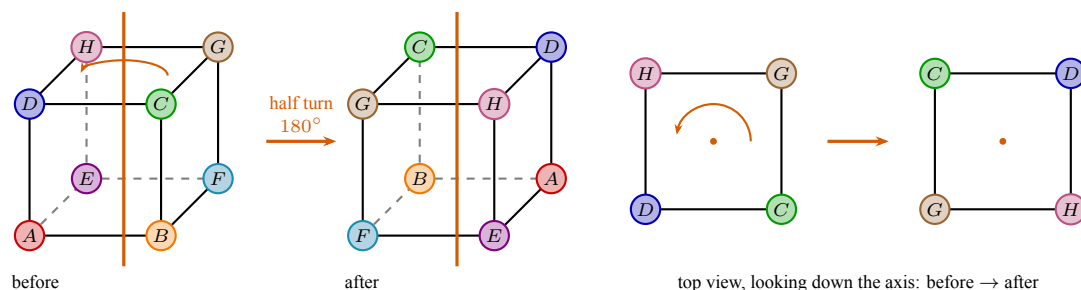


3 face axes

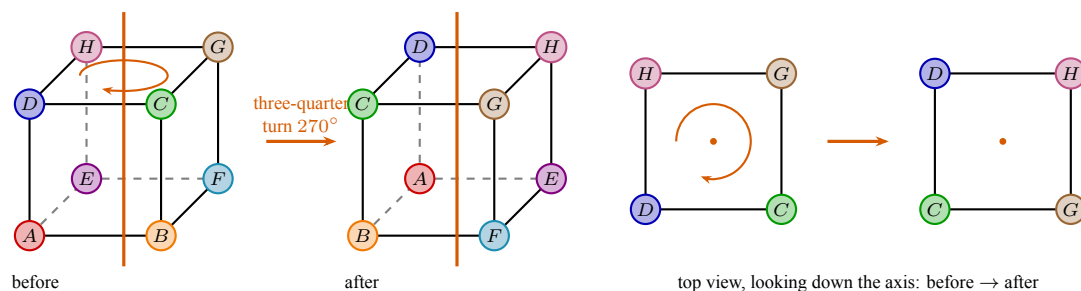
top-bottom, left-right, front-back; each joins the centres of two opposite faces. Three turns about each: $3 \cdot 3 = 9$ turns.



Quarter turn, 90° . Each corner of the bottom face moves one place round, $A \rightarrow B \rightarrow F \rightarrow E \rightarrow A$, and so does the top face, $D \rightarrow C \rightarrow G \rightarrow H \rightarrow D$. Diagonals: $d_1 \rightarrow d_2 \rightarrow d_4 \rightarrow d_3 \rightarrow d_1$, a 4-cycle. *Top view* down the face axis, from above the top face: the square $DCGH$; A, B, F, E are hidden beneath D, C, G, H , and the axis is the orange dot at the centre. The ring of four colours steps round by one quarter.



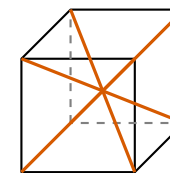
Half turn, 180° . Every corner jumps to the opposite corner of its face: $A \leftrightarrow F$, $B \leftrightarrow E$, $C \leftrightarrow H$, $D \leftrightarrow G$. Diagonals: $(d_1 d_4)(d_2 d_3)$, two disjoint swaps. *Top view* down the face axis, from above the top face: the square $DCGH$; A, B, F, E are hidden beneath D, C, G, H , and the axis is the orange dot at the centre. Each colour goes to the diametrically opposite corner of the square.



Three-quarter turn, 270° ($= -90^\circ$). The quarter turn in the other direction: every corner goes the opposite way, $A \rightarrow E \rightarrow F \rightarrow B \rightarrow A$ and $D \rightarrow H \rightarrow G \rightarrow C \rightarrow D$. Diagonals: $d_1 \rightarrow d_3 \rightarrow d_4 \rightarrow d_2 \rightarrow d_1$, the inverse 4-cycle. *Top view* down the face axis, from above the top face: the square $DCGH$; A, B, F, E are hidden beneath D, C, G, H , and the axis is the orange dot at the centre. The ring steps round by one quarter the other way; this is why the table writes $\pm 90^\circ$.

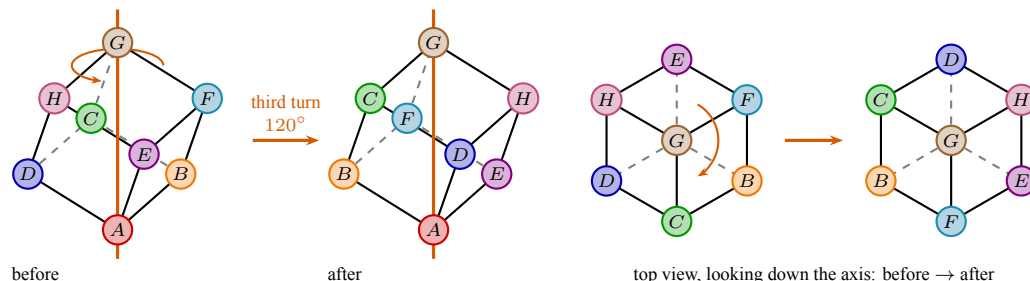
Interlude: A3. Long diagonals: two turns about each, and why not a third kind

For a slanted axis the cube is drawn standing on the corner the axis passes through, so the axis is vertical, exactly as on the previous page. Same colour code, same three views. The cube on the right shows all four long diagonals; the rows use AG , which is the diagonal d_1 . The third row shows a 60° turn, which is *not* a symmetry, to explain why the count stops at two.

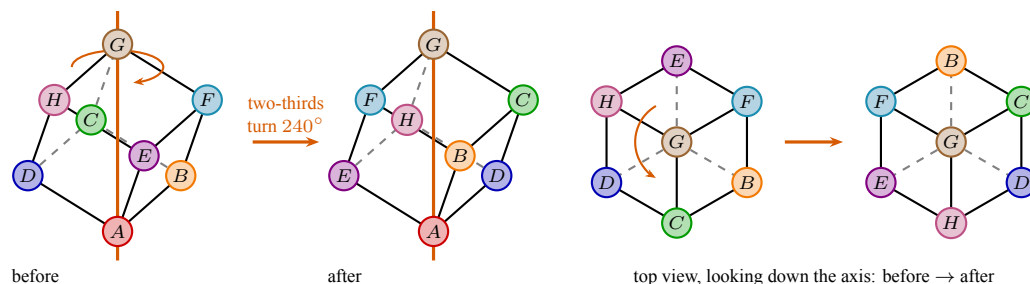


4 long diagonals

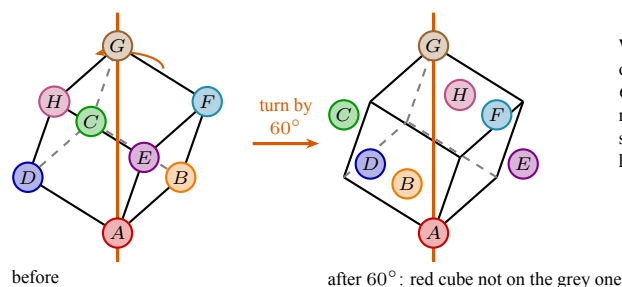
each joins two opposite corners through the centre. Two turns about each: $4 \cdot 2 = 8$ turns.



Third turn, 120° , about the long diagonal AG . The cube stands on A so the axis is vertical; A and G stay. The lower ring turns one step, $B \rightarrow D \rightarrow E \rightarrow B$, and so does the upper ring, $C \rightarrow H \rightarrow F \rightarrow C$. Diagonals: d_1 fixed, $d_2 \rightarrow d_4 \rightarrow d_3 \rightarrow d_2$, a 3-cycle. *Top view* down the diagonal AG , from above G : a regular hexagon with G at the centre and A hidden beneath it; solid spokes go to the neighbours C, F, H of G , dashed spokes to the neighbours B, D, E of A . The hexagon of colours rotates by one third.



Two-thirds turn, $240^\circ (= -120^\circ)$. The third turn the other way. Lower ring $B \rightarrow E \rightarrow D \rightarrow B$, upper ring $C \rightarrow F \rightarrow H \rightarrow C$. Diagonals: d_1 fixed, $d_2 \rightarrow d_3 \rightarrow d_4 \rightarrow d_2$, the inverse 3-cycle. *Top view* down the diagonal AG , from above G : a regular hexagon with G at the centre and A hidden beneath it; solid spokes go to the neighbours C, F, H of G , dashed spokes to the neighbours B, D, E of A . The hexagon rotates by one third the other way; this is why the table writes $\pm 120^\circ$.



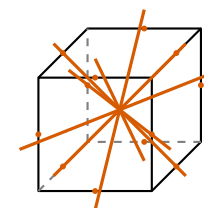
Why not 60° ? Turn the cube by only 60° and it does *not* land on itself: in the “after” picture the turned cube (red) sits over a grey ghost of the original, and no corner lands on a corner. The reason is height. Along the axis the six outer corners lie on two rings, the upper ring C, F, H one third of the way down and the lower ring B, D, E two thirds down, and around the hexagon the rings alternate. A 60° turn moves each corner one step round the hexagon, from an upper-ring spot to a lower-ring spot, at the wrong height. A 120° turn moves two steps, ring to same ring. (Seen from above, the outline would still fit after 60° , which is why the top view alone can mislead: it hides the heights.)

Watch out

Two other ways to see that only 120° and 240° work: looking straight down the axis, the three edges meeting at the near corner form a Y inside the hexagon, and the Y has only 3-fold symmetry; and a rotation about the axis must map the three-corner upper ring to itself, so only multiples of $360^\circ/3$ are possible. A 60° turn *combined* with a reflection in the plane halfway up the axis does map the cube to itself, but that is a mirror-type symmetry you cannot perform by turning the cube, so it is not among the 24 rotations; with mirrors allowed the group has 48 elements (Interlude B2).

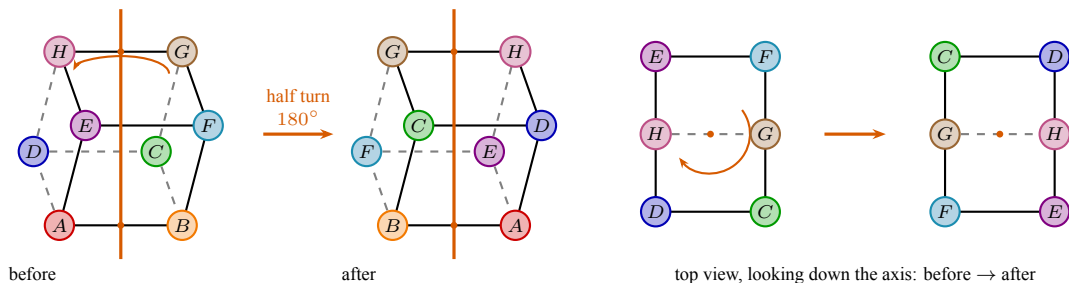
Interlude: A4. Edge axes: one turn about each

An edge axis joins the midpoints of two opposite edges. The cube is drawn standing on that edge so the axis is vertical. Same colour code, same three views. The cube on the right shows all six edge axes with the twelve edge midpoints dotted; the row uses the axis through the midpoints of AB and GH .



6 edge axes

each joins the midpoints of two opposite edges; 12 edges make 6 opposite pairs. One turn about each: 6 turns.



Half turn, 180°, about an edge axis. The axis passes through the midpoints (dots) of the opposite edges AB and GH , and the cube stands on the edge AB . Every corner swaps with a partner: $A \leftrightarrow B$, $G \leftrightarrow H$, $C \leftrightarrow E$, $D \leftrightarrow F$. Diagonals: $d_1 \leftrightarrow d_2$ only, one swap. *Top view* down the edge axis, from above GH : the $1 \times \sqrt{2}$ rectangle $CDEF$; G, H sit on the long sides with B, A hidden beneath them. The half turn sends every colour to the diametrically opposite spot, and a second half turn brings everything home.

Interlude: A5. Putting the three kinds together, and how to read the pictures

kind of axis	how many axes	turns about each	turns in all
through two opposite face centres	3	3 (90° , 180° , 270°)	9
through two opposite corners (long diagonal)	4	2 (120° , 240°)	8
through the midpoints of two opposite edges	6	1 (180°)	6
plus the identity			1
total			24

Why?

How to read a diagonal off a before/after pair: a diagonal is named by its two ends, so look where both ends went. In the quarter turn, A now sits in B 's corner and G in H 's, so the diagonal $d_1 = AG$ now occupies the slot of $BH = d_2$. Likewise B, H have gone to the corners F, D , so d_2 occupies the slot of d_4 ; and so on round the 4-cycle. In the edge half-turn, C and E have swapped, so the diagonal $CE = d_3$ occupies its own slot, reversed: as a diagonal it is fixed.

Why?

Why 270° and 240° are listed as “ -90° ” and “ -120° ”: turning by 270° one way lands the cube exactly where turning by 90° the other way does, so it is the same symmetry, the inverse of the quarter turn, with every corner move reversed. That is why the table writes $\pm 90^\circ$ and $\pm 120^\circ$ and counts two turns per axis. The half turns are their own inverses, so they come singly.

Watch out

Same kind of turn, different axis, different permutation. The quarter turn drawn is about the top–bottom axis and gives the 4-cycle $(d_1 d_2 d_4 d_3)$; the quarter turn about the front–back axis gives a different 4-cycle. There are 6 quarter turns in all (3 axes, 2 directions) and 6 four-cycles in S_4 : one each. The same matching holds row by row: 3 half turns and 3 double swaps, 8 third turns and 8 three-cycles, 6 edge turns and 6 single swaps.

Try it

Do the 120° picture and then the 240° picture one after the other, reading the “after” of the first as the “before” of the second: every colour returns home. One third plus two thirds is a full turn, and in S_4 : $(d_2 d_3 d_4)(d_2 d_4 d_3) = ()$. Then try the quarter turn followed by the edge half-turn, and find which of the 24 rows the result belongs to.

That was satisfying—we certainly understand the cube better now than we did at the start. In geometric group theory we aim to understand symmetries of much more complicated (and beautiful!) objects.

Here is the plan for this first office hour. We'll start by revisiting the basic examples of groups from any undergraduate course in abstract algebra. But we will look at these groups from the point of view that *every* group should be the collection of symmetries of some geometric object, just as the symmetric group on four letters is the collection of symmetries of a three-dimensional cube. To some extent, a lot of the material in this first office hour is a review of some of the important ideas from a first course in abstract algebra; you may want to skim through it if you are comfortable with those concepts.

- 1 In the next office hour, we will make sense of and prove the following theorem (Theorem 2.2 below):

Every group is naturally identified with the collection of symmetries of some geometric object, namely, the Cayley graph of the group.

The goal of this book is to convince you that this theorem—along with the many other theorems in this book—is the beginning of a beautiful and fruitful dictionary relating the algebraic and geometric structures arising in the wild and fascinating world of infinite groups.

1.1 GROUPS

Let's recall the definition of a group. Then we'll explain why you should think of a group in terms of symmetries.

- 2 First, a *multiplication* on a set G is a function

$$G \times G \rightarrow G.$$

- 3 The image of (g, h) is usually written gh . In other words, a multiplication is a way of combining two elements of G in order to get a third one.

Then, a *group* is a set G together with a multiplication that satisfies the following three properties:

- 4
- *Identity.* There is an element 1 of G such that $1g = g1 = g$ for all $g \in G$. The element 1 is called the *identity element* of G .
 - *Inverses.* For each $g \in G$, there is an element $h \in G$ such that $gh = hg = 1$. The element h is often denoted by g^{-1} and is called the *inverse* of g .
 - *Associativity.* The multiplication on G is associative, that is:

$$(fg)h = f(gh)$$

for all $f, g, h \in G$.

What does this abstract, formal definition have to do with symmetries? Let's first think about this in the case of the cube. We defined a symmetry of the cube as a rigid transformation of the cube. The way we multiply two symmetries g and h is:

1 Theorem 2.2 in advance

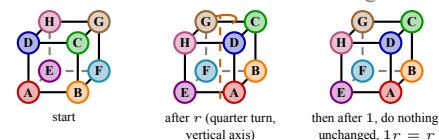
This sentence is the destination of Office Hours 1 and 2. A *Cayley graph* is a picture built from a group and a generating set: one dot per element, an arrow from g to gs for each generator s . The arrowed pictures on pages 9–11 are Cayley graphs in disguise.

2 “A multiplication on a set”

A function $G \times G \rightarrow G$ takes an *ordered pair* (g, h) and returns one element of G : the result stays in G (“closure”), and the order matters, so gh and hg may differ. For $\mathbb{Z}$ it means addition; for symmetries, composition: gh is “do h , then g ”.

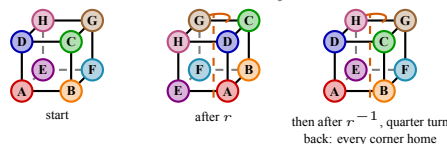
3 Identity: on the cube, “leave it alone”

The symbol 1 is a *name*, not the number one: in $\mathbb{Z}$ the identity is 0 , for the cube it is the rotation by 0° . Doing nothing after a quarter turn r leaves the picture of r untouched, and doing nothing before it gives r too, so $1r = r1 = r$. Each disc names the original corner now at that position.



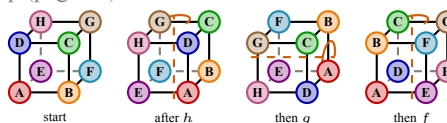
4 Inverses: on the cube, “turn it back”

g^{-1} is the move that cancels g ; it is not “ $1/g$ ” unless the group is numbers under multiplication. For the quarter turn r the inverse is the quarter turn the other way, and three more turns forward do the same, so $r^{-1} = r^3$. Undo before or after, every disc returns home: $gh = hg = 1$.



5 Associativity: brackets do not matter

fgh has one meaning. On the cube, $(fg)h$ and $f(gh)$ both put the corners through h , then g , then f ; the brackets only say which two moves you merge into a single rotation first, so both land on the last cube below (Interlude K2 draws the merged versions). Composition of functions is always associative; the free group (page 12) is where this takes real work.



Watch out

Nothing says $gh = hg$; groups where it holds are *abelian* (page 17). The cube's group is not: h then g above differs from g then h . Try it with a real cube.

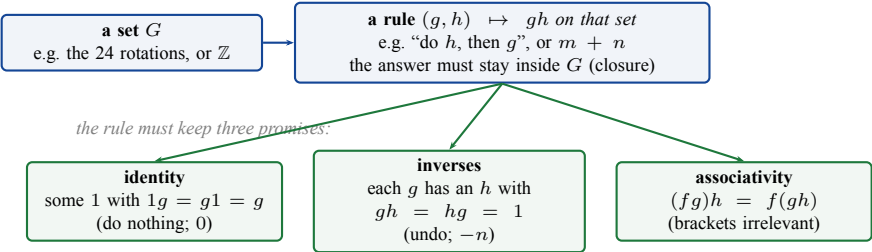
6 From the formal definition back to symmetries

The next page checks the axioms for the cube in words. Interlude K takes the definition apart, with a checklist of examples and non-examples; Interlude K2 draws each axiom on the cube.

Interlude: K. Page 5: the definition of a group, unpacked

The definition on page 5 is short, but every word of it is load-bearing. This page takes it apart: what the data are, what the three promises say, which familiar objects pass the test and which fail, and why exactly these three promises are the right ones. The next Interlude then draws each promise on the cube.

1. Anatomy of the definition



The *rule* is part of the data: the same set can be a group with one rule and fail with another ($\mathbb{Z}$ with $+$ passes, $\mathbb{Z}$ with $\times$ or $-$ fails; see the checklist). “Multiplication” is only a word for the rule; nothing about it need resemble multiplying numbers.

2. Why exactly these three promises?

They are precisely what is needed to *solve equations*. Suppose g and k are known and you want the unknown x with $gx = k$; on the cube: “which move x , followed by g , gives k ?”

$gx = k$
 $g^{-1}(gx) = g^{-1}k$
 $(g^{-1}g)x = g^{-1}k$
 $1x = g^{-1}k$
 $x = g^{-1}k$

given
apply g^{-1} on the left of both sides
regroup
 $g^{-1}g = 1$
 $1x = x$

$(inverses\ exist)$
 $(associativity)$
 $(inverses,\ both\ sides\ needed)$
 $(identity)$

Each promise is used exactly once, and the answer is unique: do k , then undo g . Drop any one axiom and this calculation breaks. That is also why the cancellation law $gx = gy \Rightarrow x = y$ holds in every group.

3. Checklist: which of these are groups?

set	rule	closed	identity	inverses	assoc.	group?
$\mathbb{Z}$	$+$	✓	0	$-n$	✓	yes
$\{0, 1, 2, \dots\}$	$+$	✓	0	✗ $(-5 \notin)$	✓	no
$\mathbb{Z}$	$-$	✓	✗ $(0 - 5 \neq 5)$	$-$	✗	no
$\mathbb{Z}$	$\times$	✓	1	✗ $(\frac{1}{2} \notin \mathbb{Z})$	✓	no
$\{1, -1\}$	$\times$	✓	1	each is its own	✓	yes
$\mathbb{R}$	$\times$	✓	1	✗ $(0\ \text{has none})$	✓	no
$\mathbb{R} \setminus \{0\}$	$\times$	✓	1	$1/x$	✓	yes
24 cube rotations	“then”	✓	do nothing	undo	✓	yes
the 6 quarter turns	“then”	✗	$-$	$-$	$-$	no
$\{1, 3\}$ face half turns	“then”	✓	1	each its own	✓	yes

For $\mathbb{Z}$ with $-$: $5 - 0 = 5$ but $0 - 5 \neq 5$, so 0 is an identity on one side only, and $(8 - 4) - 2 \neq 8 - (4 - 2)$. Associativity never fails for symmetries, because “do one, then the other” is composition of functions.

4. Closure can fail inside the cube

The six quarter turns look like a natural family, but two of them in a row need not be a quarter turn, so the family is not closed under “then” and is not a group by itself:

The last cube is the 120° corner turn about the diagonal CE (compare the dictionary in Interlude B2). The full set of 24 rotations *is* closed: two rotations in a row are a rotation, because they are exactly the moves you can perform with the cube in your hands.

Try it

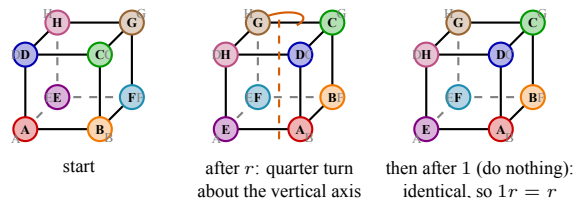
Decide, using the checklist columns: the even integers with $+$; the odd integers with $+$; the 8 corner turns together with “do nothing” (hint: try two corner turns about different diagonals); the four rotations about a single face axis. Then solve $gx = k$ on the cube with g a quarter turn about the vertical axis and k the half turn about the same axis.

Interlude: K2. Page 5: the three axioms, each checked on the cube

The 24 rotations of the cube are the book's model group, so each axiom is tested on the cube itself, with the integers under addition as a one-line second example. Reading the pictures: grey letters are the fixed *positions* $A, \dots, H$; the coloured disc at a position names the *corner of the original cube* now sitting there, so a disc away from its grey letter has moved. An orange dashed line is the axis of the move just made; “multiply” means “do the right-hand move, then the left-hand one” (page 6).

Identity: a move that changes nothing

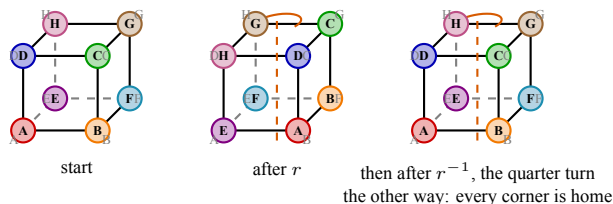
“Leave the cube alone” is a rotation (by 0°), and it is the identity: doing it after any move r leaves the picture of r untouched, and doing it before r gives r as well. So $1r = r1 = r$.



Integers: the identity is 0, not 1: $0 + 5 = 5 + 0 = 5$, while $1 + 5 \neq 5$.

Inverses: every move can be undone

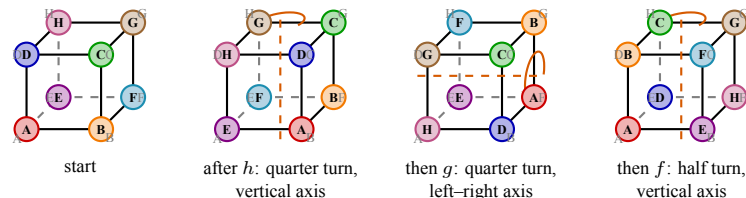
The inverse of the quarter turn r is the quarter turn the other way about the same axis; three more turns in the original direction do the same job, so $r^{-1} = r^3$. Undoing before or after both return every disc to its grey letter: $r^{-1}r = rr^{-1} = 1$.



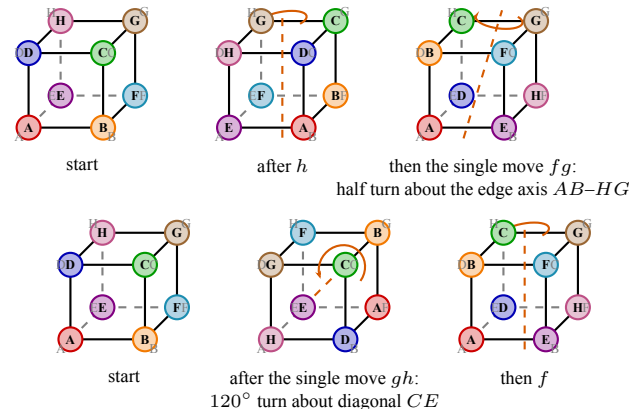
Integers: the inverse of 5 is -5 , since $5 + (-5) = (-5) + 5 = 0$; it is not $1/5$, which cancels 5 under the wrong operation. Half turns are their own inverses, like 0 in $\mathbb{Z}$.

Associativity: brackets do not matter

Take three moves: h a quarter turn about the vertical axis, g a quarter turn about the left-right axis, f a half turn about the vertical axis. Performed one after another they give this:



$(fg)h$ means: do h , then the *single* rotation that fg amounts to. $f(gh)$ means: do the single rotation gh , then f . Both end at the same cube as the row above:



Brackets only choose which two moves you *merge into one* first; the cube still experiences h , then g , then f , so the final position cannot depend on them. This is why every group of symmetries is associative for free: composing functions is associative.

Integers: $(2 + 5) + 3 = 10 = 2 + (5 + 3)$. Subtraction is *not* associative, $(8 - 4) - 2 = 2$ but $8 - (4 - 2) = 6$, which is one reason $\mathbb{Z}$ with subtraction is not a group.

Try it

Use a real cube or the dictionary of Interlude B2: what is the inverse of a 120° corner turn? Of an edge half-turn? Check on the pictures above that fg really is a half turn about the edge axis through the midpoints of AB and HG , by following the discs A and B . Then explain why $\{0, 1, 2, \dots\}$ with addition satisfies identity and associativity but fails inverses.

Big picture

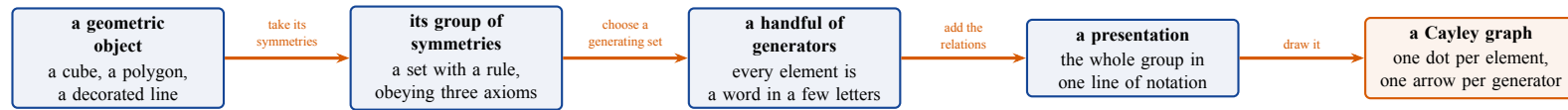
The three axioms, illustrated on the cube: what to take from this page. Each axiom is checked with before/after cube pictures. Grey letters mark fixed positions, coloured discs show which original corner sits there, and an orange dashed axis marks the move just made.

- *Identity:* start, after a quarter turn r , then after “do nothing”. The last two cubes are identical, so $1r = r1 = r$.
- *Inverses:* start, after r , then after the quarter turn the other way. Every disc is back on its grey letter, so $r^{-1}r = rr^{-1} = 1$, and $r^{-1} = r^3$.
- *Associativity:* three moves h, g, f shown one after another, then two more rows. One row does h and then the single rotation fg , which turns out to be a half turn about the edge axis $AB-HG$. The other does the single rotation gh , a 120° turn about the diagonal CE , and then f . All three rows end at the same cube.

Each axiom also gets a one-line integer example, with 0 as the identity, -5 as the inverse of 5, and subtraction as a non-associative contrast. The try-it box asks for the inverses of a corner turn and an edge half-turn.

Interlude: O. What each idea in this office hour actually buys you

Each definition in this office hour looks like bookkeeping until you see what becomes possible with it. This page is the answer to “why are we doing this”, concept by concept, with the smallest convincing example in each case. The chapter as a whole is one long pipeline.



Office Hour 1 walks left to right. Office Hour 2 closes the loop by proving that the symmetries of the graph on the right are exactly the group in the middle.

Symmetry as a rigid motion

The gap it fills. “Symmetry” is a feeling until you say exactly which motions count.

What it buys. A finite, countable, composable collection instead of a vague impression.

Smallest example. The cube has 24 rotations, and you can check that number two independent ways: 8 corners times 3 turns, and $9 + 8 + 6 + 1$ by kind of axis. Two different countings agreeing is evidence the definition is the right one.

The three axioms

The gap it fills. Which collections deserve the name, and which facts hold for all of them at once.

What it buys. Everything provable without knowing a single element: the identity is unique, inverses are unique, $gx = k$ has exactly one solution, and $gx = gy$ forces $x = y$.

Smallest example. The six quarter turns of the cube are a natural-looking family, and they fail: two of them in a row give a corner turn. Without closure there is no group and none of the above applies.

Isomorphism

The gap it fills. Two groups built from unrelated material can be the same group.

What it buys. A dictionary. Every question about one becomes a question about the other, and you may work in whichever is easier.

Smallest example. The rotations of the cube are S_4 . “Which turn is this?” becomes “which permutation of four things is this?”, and the second question needs no spatial imagination at all.

Cycle notation

The gap it fills. A permutation written as a list of images is unreadable and hopeless to multiply.

What it buys. A short normal form, and a multiplication you can carry out with a pencil in a few seconds.

Smallest example. $(26) \cdot (124)(35) = (1624)(35)$, computed by pushing six numbers through two factors. Try describing the same product as two motions of a cube and the contrast is immediate.

Generating sets

The gap it fills. A group can be enormous or infinite, so listing it is not an option.

What it buys. The whole group described by a handful of elements, and a homomorphism out of it determined by where those few elements go.

Smallest example. Two reflections generate all ten symmetries of the pentagon; $n - 1$ neighbour swaps generate all $n!$ permutations. In both cases a small list controls everything.

Relations and presentations

The gap it fills. Words in the generators are not unique, so you need to say which words are equal.

What it buys. An entire group compressed into one line, and a mechanical way to decide equality in favourable cases.

Smallest example. $\langle a, b \mid aba^{-1}b^{-1} \rangle$ is $\mathbb{Z}^2$. Delete that single relator and the same two generators give F_2 , a completely different group. One line of notation is doing all the work.

The free group

The gap it fills. What if you impose no relations at all?

What it buys. A universal source: every group is a free group with relations added, so free groups sit underneath the whole subject.

Smallest example. Within distance k of the identity, $\mathbb{Z}^2$ has about k^2 elements and F_2 about 3^k . Same number of generators, utterly different sizes, and the only difference is one relation.

Homomorphism, kernel, quotient

The gap it fills. Comparing two groups, and building smaller ones from larger ones.

What it buys. Structure-preserving maps, the first isomorphism theorem, and the meaning of the slash in $\mathbb{Z}/n\mathbb{Z}$.

Smallest example. The sign map $S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ turns a question about rearranging n objects into a question about even and odd, which is the difference between a hard problem and an easy one.

The Cayley graph

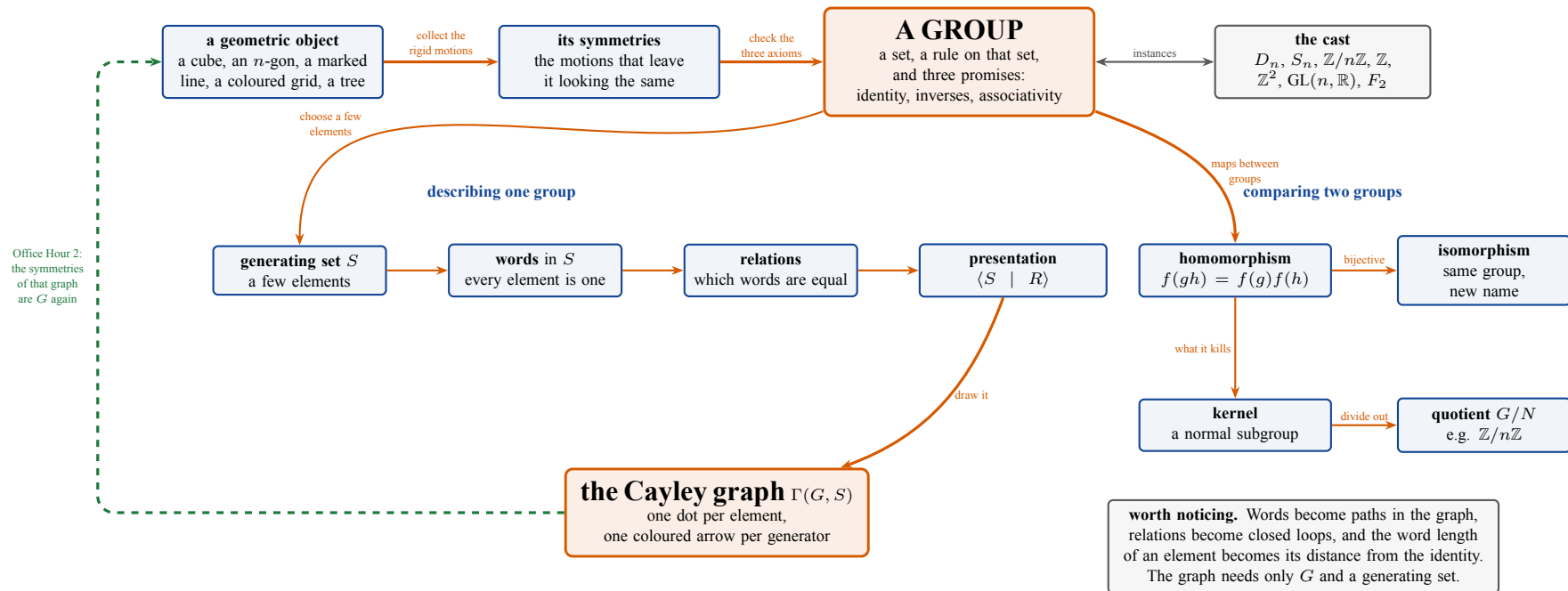
The gap it fills. Algebra by itself gives nothing to look at.

What it buys. One dot per element, one coloured arrow per generator, and a picture in which words become paths and relations become loops.

Smallest example. The arrowed polygon, the arrowed line, the coloured grid and the four-valent tree are the same construction applied to $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}$, $\mathbb{Z}^2$ and F_2 . All four appeared in this office hour before the name did.

Interlude: Q. The map of this office hour: how the ideas fit together

Everything in Office Hour 1 is one journey with a loop at the end. You start with something you can hold, extract its symmetries, notice that the result obeys three rules, and then spend the rest of the chapter learning to describe, compare and finally redraw such a thing. Read the map along the orange arrows.



Why?

Why the loop at the bottom matters. The journey begins with a geometric object and produces a group. The dashed arrow says the journey can be run backwards: from the group alone you can build a graph, and the symmetries of that graph turn out to be the group you started from. That closes the circle, and it is the theorem Office Hour 2 proves. Until it is proved, “a group is a collection of symmetries” is only a slogan supported by examples.

Big picture

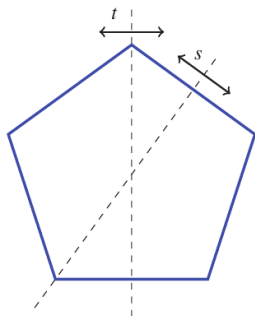
The two clusters in the middle answer two different questions and it is worth keeping them apart. The left cluster asks *how do I write this group down*, and its answer is a presentation. The right cluster asks *how does this group compare with that one*, and its answers are the isomorphism theorems. The examples along the top are where both clusters are tested: every one of them appears with a generating set, a presentation, and at least one homomorphism.

- 1 “do h , then do g .” This is just like composing functions: in the composition $g \circ h$ we apply h first.

We now see that the set of symmetries of the cube satisfies the definition of a group. The identity symmetry 1 is the rigid transformation of the cube that leaves it alone. Indeed, doing one rigid transformation g and then doing 1 is the same as doing g . The inverse of a symmetry g is the symmetry that undoes g . Associativity follows from the associativity of composition of functions: doing h , then g and f is the same as doing h and g , then f .

As we already said, we’d like to eventually convince you that every group—that is, every set with a multiplication as above—is the set of symmetries of some object. First, let’s look at a few examples of groups and explain how they each can be thought of as a set of symmetries.

- 2 **The dihedral group.** The dihedral group D_n is the set of symmetries (rigid transformations) of a regular n -gon in the plane. The multiplication is the same as in the example of the cube. Two symmetries of the regular pentagon are the reflections s and t shown here:



For general n we can take s to be the reflection through any line that passes through the center of the n -gon and a vertex of the n -gon, and t to be the reflection about the line that differs from the first line by a π/n rotation. The elements s and t generate the group D_n , by which we mean that every symmetry of D_n is obtained by multiplying, s , t , s^{-1} , and t^{-1} (actually, in this example $s = s^{-1}$ and $t = t^{-1}$).

- 3 generate the group D_n , by which we mean that every symmetry of D_n is obtained by multiplying, s , t , s^{-1} , and t^{-1} (actually, in this example $s = s^{-1}$ and $t = t^{-1}$).

- 4 For example, the clockwise rotation by m “clicks” is $(st)^m$.

Adam Piggott’s Office Hour 13 on Coxeter groups discusses examples of groups that are all based on the idea of the dihedral groups.

- 5 **The symmetric group.** The symmetric group S_n is the set of permutations of the set $\{1, \dots, n\}$, with multiplication given by composition of functions. This makes sense since a permutation of $\{1, \dots, n\}$ is really just a bijective function $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$. If we want to think of S_n as the set of symmetries of some object, we can think of it as the symmetries of a set of n points.

- 1 “Do h , then do g ” is gh

This right-to-left convention comes from writing functions on the left of their inputs: $(g \circ h)(x) = g(h(x))$, so h acts first. The computations on page 7 depend on it.

- 2 **The dihedral group D_n**

Now reflections *are* allowed: a flat polygon lives in the plane, so you can lift it and flip it over. Before the new vocabulary, run the three checks. Identity: leave the polygon alone. Inverses: turn a rotation back, and a reflection undoes itself. Associativity: free, because symmetries are functions and “do h , then g ” is composition. Only then the extra facts: D_n has $2n$ elements, n rotations (multiples of $2\pi/n$, including the identity) and n reflections; it is not abelian; the rotations form a subgroup. (Some books call it D_{2n} , naming it by its size; this book names it by the polygon.)

Looking ahead

Interlude L runs all three checks with before/after pictures of the pentagon, then lists the extra properties; Interlude D turns the ten symmetries into words in s and t .

In the book’s figure t is the reflection in the vertical axis and s the one in the axis rotated by $\pi/5 = 36^\circ$ from it. For odd n every axis runs through a vertex and the midpoint of the opposite edge.

- 3 “ s and t generate D_n ”

A set S generates G if every element of G is a product of elements of S and their inverses. This idea runs through the whole book; it is made formal in §1.4.

Why?

Reflecting in two lines that meet at angle θ is the same as rotating by 2θ about their intersection. Here $\theta = \pi/n$, so st is the rotation by $2\pi/n$: one “click”. Hence $(st)^m$ is m clicks, all n rotations are $(st)^m$ for $m = 0, \dots, n-1$, and every reflection is (rotation)· s . So s, t really do generate everything.

- 4 $s = s^{-1}$

Reflecting twice returns everything to its place, so $s^2 = 1$: s is its own inverse.

- 5 **The symmetric group S_n**

The same three checks, now with no geometry to respect. Identity: leave every point where it is. Inverses: read the cycle backwards, or reverse every arrow. Associativity: free again, since the rule is composition of functions. So *every* rearrangement counts, and $|S_n| = n!$; “symmetries of n points” just means there are no distances or shape to preserve. Cube = S_4 says the rotations permute the four diagonals as freely as four loose points.

Looking ahead

Interlude L checks the axioms for S_n with tokens moving between slots, and collects what makes it interesting: size $n!$, not abelian, generated by neighbour swaps.

Interlude: L. Page 6: the two new examples, first checked against the three axioms

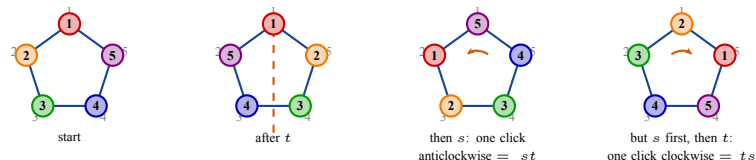
The book introduces D_n and S_n in a sentence each. It is worth doing slowly once: say what the *set* is and what the *rule* is, check the three promises, and only then collect the extra features. In the pictures, grey numbers are fixed positions or slots and a coloured disc names the original corner or token now sitting there.

The dihedral group D_5 (the pentagon)

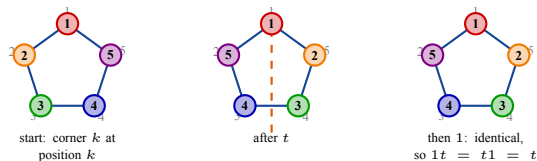
Set: the 10 rigid motions of a regular pentagon, 5 rotations (by $0^\circ, 72^\circ, \dots, 288^\circ$) and 5 reflections.

Rule: gh means “do h , then g ”, as for the cube. Reflections count here because a flat polygon can be flipped over in space.

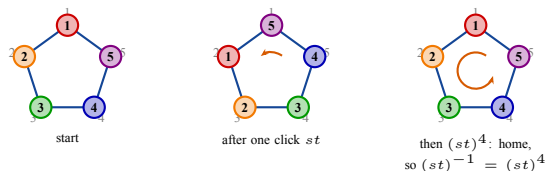
0. Closure. Two reflections in a row are not a reflection, but they are still a symmetry, so nothing leaves the set. The same picture shows the order matters.



1. Identity. “Leave the pentagon alone” is the rotation by 0° ; before or after any move it changes nothing.



2. Inverses. A rotation is undone by turning back, which is the same as turning forward the rest of the way. A reflection is its own inverse: apply t twice and the second picture above is undone, $t^2 = 1$.



3. Associativity. Free of charge: each symmetry is a function from the pentagon to itself, and $((fg)h)(x) = f(g(h(x))) = (f(gh))(x)$ for every point x . So a word like tst has one meaning.

Big picture

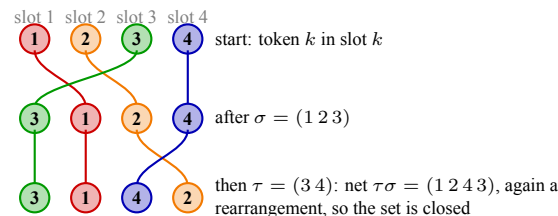
Now the extra properties. $|D_n| = 2n$. Not abelian for $n \geq 3$: the first picture shows $st \neq ts$. The reflections s, t generate it, since st is one click: $(st)^m$ gives all n rotations and $s(st)^m$ all n reflections (Interlude D lists the ten words). The rotations alone form a subgroup, a copy of $\mathbb{Z}/n\mathbb{Z}$. The relations $s^2 = t^2 = 1$ and $(st)^n = 1$ pin the group down completely, which is the Coxeter idea named in the last line of the page.

The symmetric group S_n (permutations)

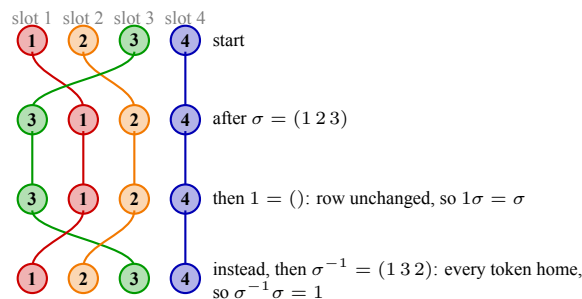
Set: all bijections $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$, that is, all ways of rearranging n tokens among n slots.

Rule: composition, $\tau\sigma$ meaning “do σ , then τ ”. Nothing geometric has to be respected, which is why *every* rearrangement is allowed.

0. Closure. A rearrangement followed by a rearrangement is a rearrangement: every slot still ends up with exactly one token.



1. Identity and 2. inverses, in one picture. The identity is the permutation fixing every slot, written 1 or $()$. The inverse is the cycle read backwards, or the loop picture of page 7 with every arrow reversed; the tokens retrace their strands.



3. Associativity. Again free: for any x , both $(fg)h$ and $f(gh)$ send x to $f(g(h(x)))$. That is why the book can say “multiplication is composition of functions” and stop. Only groups given by a *formula*, like $\mathbb{Z}/n\mathbb{Z}$ on page 8, or by *words*, like the free group on page 12, make you work for this axiom.

Big picture

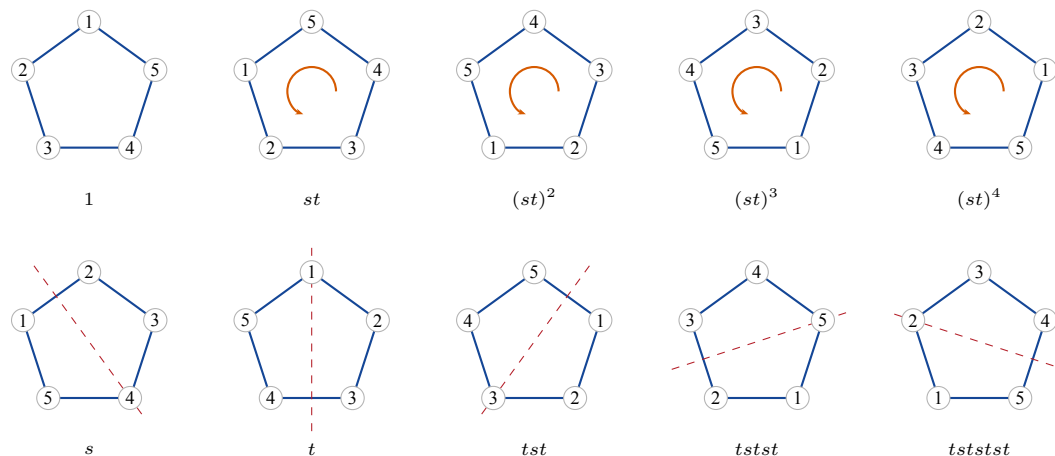
Now the extra properties. $|S_n| = n!$: the token in slot 1 has n choices, the next $n-1$, and so on. Not abelian for $n \geq 3$, since $(12)(23) = (123)$ but $(23)(12) = (132)$. It is generated by the neighbour swaps $(12), (23), \dots$, which is exactly what the strand diagrams of page 8 draw. And S_4 is the rotation group of the cube (page 4, Interlude B2), so this abstract group has a geometric life too.

Try it

For D_5 : which word is the reflection whose axis passes through position 3? Do the five rotations alone satisfy all three axioms? Do the five reflections alone? For S_4 : compose (1234) with itself, and invert $(12)(34)$.

Interlude: D. Page 6 in a table: the ten symmetries of the pentagon as words in s and t

Take the pentagon with corners numbered 1 to 5 anticlockwise from the top, t = reflection in the vertical axis (through corner 1) and s = reflection in the axis rotated from it by $\pi/5 = 36^\circ$ (through the midpoint of edge 12 and corner 4), as in the book's figure. Every symmetry of the pentagon is one of the ten words below, and the picture shows where each word sends the numbered corners. Products are read right to left: st means “do t , then s ”.



Top row: the five rotations, $(st)^m = m$ clicks of 72° (orange arrow). Bottom row: the five reflections, with their axes dashed in red. A circled number shows where that corner has been sent.

word in s, t	type	what it does	corners 1 2 3 4 5 go to
1	rotation	by 0° (0 clicks)	1 2 3 4 5
st	rotation	by 72° (1 click)	2 3 4 5 1
$(st)^2$	rotation	by 144° (2 clicks)	3 4 5 1 2
$(st)^3$	rotation	by 216° (3 clicks)	4 5 1 2 3
$(st)^4$	rotation	by 288° (4 clicks)	5 1 2 3 4
s	reflection	axis at 126° from horizontal	2 1 5 4 3
t	reflection	axis at 90° from horizontal	1 5 4 3 2
tst	reflection	axis at 54° from horizontal	5 4 3 2 1
$tstst$	reflection	axis at 18° from horizontal	4 3 2 1 5
$tststst$	reflection	axis at -18° from horizontal	3 2 1 5 4

Why?

Why st is one click: reflecting in two lines that meet at angle θ is the same as rotating by 2θ about their crossing point. The axes of s and t meet at 36° , so st is the rotation by 72° . Then $(st)^2$ is two clicks, $(st)^3$ three, and $(st)^5 = 1$ because five clicks is a full turn.

Why?

Why the reflections are $s(st)^k$: a reflection followed by a rotation is again a reflection, in a different axis. Since $s \cdot s = 1$, the word $s(st)^k$ simplifies to $t(st)^{k-1}$, which is how the bottom row is written. The five axes are 36° apart, one per corner.

Watch out

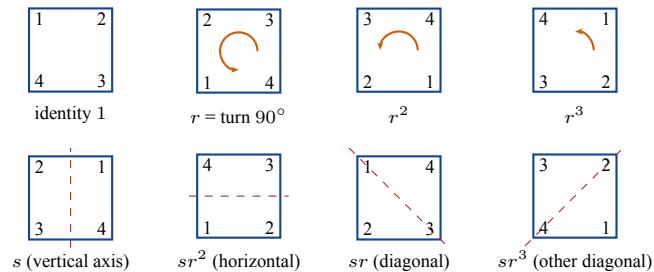
“Generate” does not mean every element *is* s or t ; it means every element is a product of s 's and t 's (inverses are not needed here, because $s^{-1} = s$ and $t^{-1} = t$). Check: the ten words above give ten different arrangements of the corners, and $|D_5| = 2 \cdot 5 = 10$, so nothing is missing and nothing is repeated.

Try it

Multiply two entries of the table by composing their corner maps, right factor first. For instance $t \cdot s$: s sends 1 2 3 4 5 to 21543, then t sends those to ... You should get the rotation by -72° , i.e. $(st)^4$: reversing the order of two reflections reverses the rotation, which is $ts = (st)^{-1}$.

Interlude: V1. Seeing a group: the square, the triangle, and the cube's axes

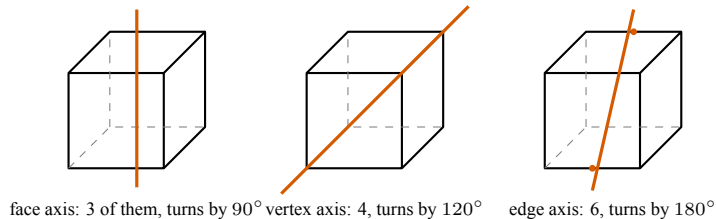
The eight symmetries of a square (D_4). Corners are numbered 1, 2, 3, 4; each picture shows where the numbers end up. r is a quarter turn, s a reflection; the other three reflections are sr , sr^2 , sr^3 (do a rotation, then s).



Try it

Compose two pictures: do r then s (find where each number goes in two steps). You should land on one of the four reflections. Then do s then r : a *different* reflection. That is $sr \neq rs$, the non-commutativity of page 5.

Three kinds of rotation axis of the cube (page 4).



The multiplication table of D_3 , coloured. Rotations are blue, reflections red; the entry in row x , column y is xy , with $r^3 = 1$, $s^2 = 1$, $rs = sr^2$.

row · column:	1	r	r^2	s	sr	sr^2
1	1	r	r^2	s	sr	sr^2
r	r	r^2	1	sr^2	s	sr
r^2	r^2	1	r	sr	sr^2	s
s	s	sr	sr^2	1	r	r^2
sr	sr	sr^2	s	r^2	1	r
sr^2	sr^2	s	sr	r	r^2	1

Three things to see at once:

- every row and every column contains each element exactly once (because $x \mapsto xy$ is a bijection: it has inverse $x \mapsto xy^{-1}$);
- the table is not symmetric about the diagonal: D_3 is not abelian;
- the four coloured blocks say “rotation·rotation = rotation, reflection·reflection = rotation, mixed = reflection”. That is the homomorphism $D_3 \rightarrow \mathbb{Z}/2\mathbb{Z}$ of page 15, and the blue block is its kernel. The 2×2 block pattern is the multiplication table of $\mathbb{Z}/2\mathbb{Z}$: a picture of $D_3/\{\text{rotations}\} \cong \mathbb{Z}/2\mathbb{Z}$.

Looking ahead

D_3 and S_3 are the same group: the six symmetries of the triangle permute its three corners in all $3! = 6$ ways. Compare the table with what you get for S_3 by composing permutations.

- 1 A *cycle* in S_n is a permutation that can be described as $i_1 \mapsto i_2 \mapsto \dots \mapsto i_k \mapsto i_1$ (and all other elements of $\{1, \dots, n\}$ stay where they are). We write this cycle as $(i_1 i_2 \dots i_{k-1} i_k)$. For instance, the cycle $(1\ 2\ 4)$ in S_6 sends 1 to 2, 2 to 4, and 4 to 1, while sending 3, 5, and 6 to themselves.
- 2 Every element of S_n is a product of disjoint cycles (disjoint means that each element of $\{1, \dots, n\}$ appears in at most one cycle). For instance, in S_6 the element

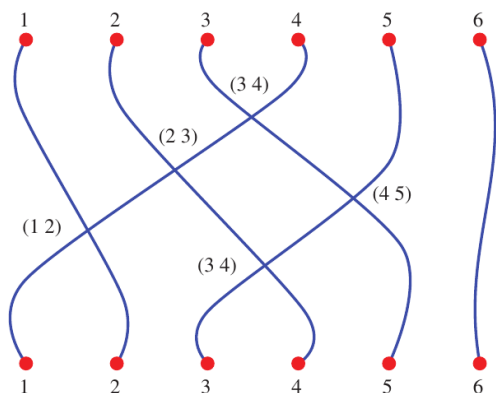
$$(1\ 2\ 4)(3\ 5)$$

is the product of the disjoint cycles $(1\ 2\ 4)$ and $(3\ 5)$. This is the permutation that sends 1 to 2, 2 to 4, and 4 to 1, and also 3 to 5 and 5 to 3, and finally 6 to itself.

As we said, the multiplication is function composition. Say, for example, we want to multiply $(1\ 2\ 4)(3\ 5)$ by $(2\ 6)$:

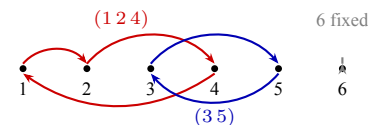
$$(2\ 6) \cdot (1\ 2\ 4)(3\ 5).$$

- 3 Where does this product send 1? Well, the first permutation (on the right) sends 1 to 2, and the second permutation sends 2 to 6, so the product (= composition) sends 1 to 6. You can use the same procedure to determine where this product sends 2, 3, 4, 5, and 6. You will find that $(2\ 6) \cdot (1\ 2\ 4)(3\ 5) = (1\ 6\ 2\ 4)(3\ 5)$.
- 4 A *transposition* is a cycle of length 2, for instance, $(3\ 5)$. It is an important fact that S_n is generated by transpositions of the form $(i\ i+1)$, where $1 \leq i \leq n-1$ (we actually used this fact implicitly in our discussion of the cube at the beginning).
- 5 Let us check that we can write $(1\ 2\ 4)(3\ 5)$ as a product of such elements in S_6 . We can draw a diagram of this permutation as follows:

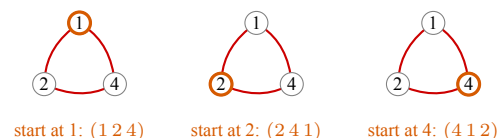


1 Cycle notation, as a picture

$(1\ 2\ 4)$ says: 1 goes to 2, 2 goes to 4, 4 goes back to 1; anything not listed stays put. Drawn as arrows on the six numbers, a cycle is a closed loop, and $(1\ 2\ 4)(3\ 5)$ is two loops with 6 left alone:

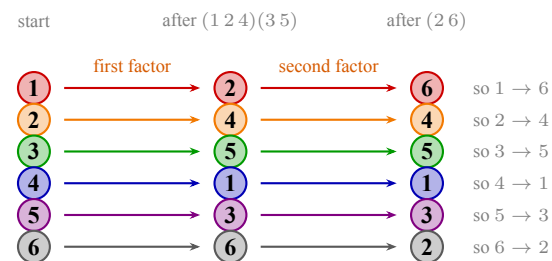


Why $(1\ 2\ 4) = (2\ 4\ 1) = (4\ 1\ 2)$: the name just lists the loop from a chosen starting number, going round once. Different starting points, same loop, same permutation:



2 Follow every number through the product

A product is a composition: the *right* factor acts first. Take each number in turn and push it through the two factors, first $(1\ 2\ 4)(3\ 5)$, then $(2\ 6)$. Each row below is one number's journey:



The first row is the book's example: 1 goes to 2 under the first factor, and 2 goes to 6 under the second, so the product sends $1 \rightarrow 6$. The right-hand column collects all six results. Now chain them into loops: $1 \rightarrow 6, 6 \rightarrow 2, 2 \rightarrow 4, 4 \rightarrow 1$ closes up as the cycle $(1\ 6\ 2\ 4)$; $3 \rightarrow 5, 5 \rightarrow 3$ is $(3\ 5)$. So $(2\ 6) \cdot (1\ 2\ 4)(3\ 5) = (1\ 6\ 2\ 4)(3\ 5)$.

Watch out

Push the numbers through the factors in the other order (left factor first) and the first row becomes $1 \rightarrow 1 \rightarrow 2$: you would get $(1\ 2\ 4\ 6)(3\ 5)$, a different element. The convention is fixed on page 6: gh means "*h*, then *g*".

4 Transpositions

A transposition swaps two numbers and fixes the rest ($(3\ 5)$ is the blue loop above); an *adjacent* one swaps two neighbours, $(1\ 2), \dots, (5\ 6)$, and the claim is that these five generate all of S_6 .

5 The diagram: one crossing = one neighbour swap

Strand i ends at $\sigma(i)$, and each crossing swaps two neighbouring positions, so five crossings mean five adjacent transpositions. Interlude C (next spread) works the product through in two ways; Interlude C2 does the same for the diagram.

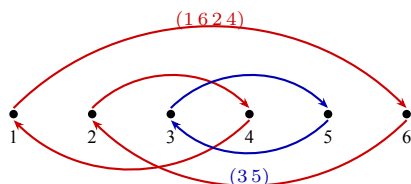
Interlude: C. Page 7, part 1: the product $(2\ 6) \cdot (1\ 2\ 4)(3\ 5)$

One rule governs everything on this page: a product of permutations is a composition of functions, and the factor on the *right* acts first. Two pictures are used. Six *slots* are fixed in a row and six coloured numbers start with k in slot k ; a permutation σ moves whatever is in slot i to slot $\sigma(i)$, and the colours travel with the numbers. A *loop picture* draws a permutation as arrows on the numbers, so each cycle is a closed loop (arrows above the line go right, arrows below go left) and a fixed number is a little loop to itself.

1. Push each number through the two factors. Cycle notation: $(1\ 2\ 4)$ means $1 \rightarrow 2$, $2 \rightarrow 4$, $4 \rightarrow 1$; $(3\ 5)$ means $3 \rightarrow 5$, $5 \rightarrow 3$; anything not mentioned stays put. First $(1\ 2\ 4)(3\ 5)$, then $(2\ 6)$:

number	after $(1\ 2\ 4)(3\ 5)$	then after $(2\ 6)$	so the product sends
1	2	6	$1 \rightarrow 6$
2	4	4	$2 \rightarrow 4$
3	5	5	$3 \rightarrow 5$
4	1	1	$4 \rightarrow 1$
5	3	3	$5 \rightarrow 3$
6	6	2	$6 \rightarrow 2$

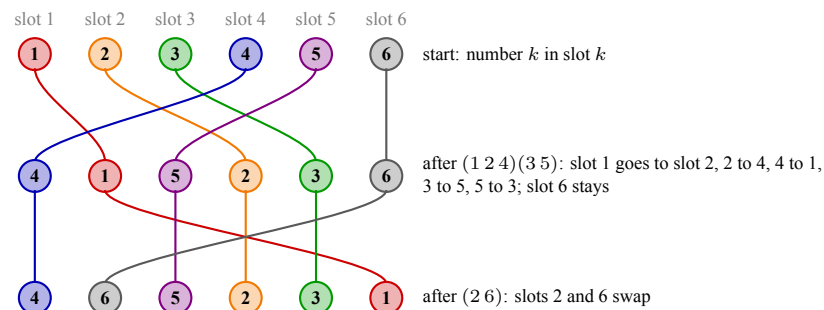
2. Write the answer as cycles. Start at 1 and follow the arrows until you return: $1 \rightarrow 6 \rightarrow 2 \rightarrow 4 \rightarrow 1$, the cycle $(1\ 6\ 2\ 4)$. The smallest unused number is 3: $3 \rightarrow 5 \rightarrow 3$, the cycle $(3\ 5)$. Everything is accounted for, so the product is $(1\ 6\ 2\ 4)(3\ 5)$. As a loop picture:



Watch out

Apply $(2\ 6)$ first instead and the first row of the table becomes $1 \rightarrow 1 \rightarrow 2$; the answer changes to $(1\ 2\ 4\ 6)(3\ 5)$, a different element. The order is not a matter of taste: it is the convention “ gh means h , then g ” fixed on page 6.

3. The same computation with the numbers moving between slots.



Read the last row: number 1 is in slot 6, 6 in slot 2, 2 in slot 4, 4 in slot 1, 3 in slot 5, 5 in slot 3. That is the right-hand column of the table, and the loops $(1\ 6\ 2\ 4)(3\ 5)$ again. The strands between the rows show each number's journey; where a strand from slot i ends at slot j , the permutation sends i to j .

Why?

Why the two pictures agree: the table follows one *number* at a time through both factors; the slot picture follows all six at once, one *factor* at a time. Each row of the slot picture is a full “where is everybody now” snapshot; the table is the same information read column by column.

Interlude: C2. Page 7, part 2: the diagram, or $(1\ 2\ 4)(3\ 5)$ as five neighbour swaps

A *transposition* swaps two numbers and fixes the rest. An *adjacent* transposition swaps two neighbours: in S_6 they are $(1\ 2)$, $(2\ 3)$, $(3\ 4)$, $(4\ 5)$, $(5\ 6)$. The claim on page 7 is that every permutation is a product of these five simple swaps, and the diagram is the proof for $(1\ 2\ 4)(3\ 5)$.

1. How to read the diagram.

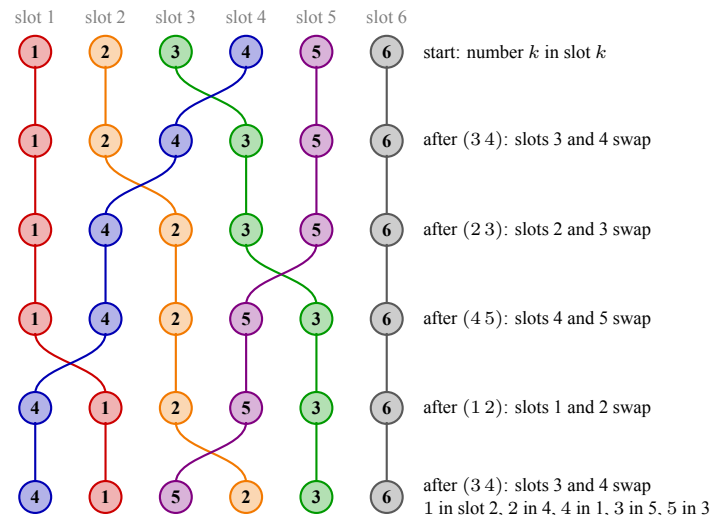
- Six strands run from the top row of labels to the bottom row. The strand starting at position i on top ends at position $\sigma(i)$ at the bottom, where $\sigma = (1\ 2\ 4)(3\ 5)$: from 1 to 2, from 2 to 4, from 4 to 1, from 3 to 5, from 5 to 3, and the strand from 6 goes straight down.
- Each crossing is where two strands trade places. At that moment they occupy two *neighbouring* positions, so the crossing is an adjacent transposition; the label names which positions.
- The crossings are drawn at five different heights on purpose. Top to bottom they read $(3\ 4)$, $(2\ 3)$, $(4\ 5)$, $(1\ 2)$, $(3\ 4)$.
- The top crossing happens first. Since the first-applied factor is written on the right, the top-to-bottom list is written right to left: $(1\ 2\ 4)(3\ 5) = (3\ 4)(1\ 2)(4\ 5)(2\ 3)(3\ 4)$.

2. Check by tracking one number, right factor first. Take 1: $(3\ 4)$ leaves it at 1; $(2\ 3)$ leaves it; $(4\ 5)$ leaves it; $(1\ 2)$ moves it to 2; $(3\ 4)$ leaves it. So $1 \rightarrow 2$, as σ says. Take 2: $(3\ 4)$ leaves it; $(2\ 3)$ moves it to 3; $(4\ 5)$ leaves it; $(1\ 2)$ leaves it; $(3\ 4)$ moves it to 4. So $2 \rightarrow 4$. The same for 3, 4, 5, 6 gives $3 \rightarrow 5$, $4 \rightarrow 1$, $5 \rightarrow 3$, $6 \rightarrow 6$.

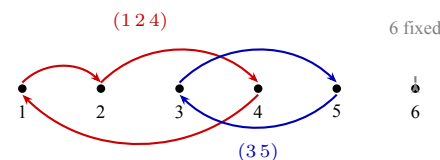
Watch out

Two things that cause most of the confusion. (a) *Right-to-left order*: “top crossing first” and “right factor first” say the same thing, so the equation looks reversed compared with the picture; it is the convention, not a reversal. (b) *Positions versus labels*: a crossing labelled $(3\ 4)$ swaps whichever two numbers currently sit in slots 3 and 4, not the numbers 3 and 4. That is why $(3\ 4)$ appears twice and moves different numbers each time (first 3 and 4, later 2 and 5).

3. The diagram with the numbers made visible. One row per height, one neighbour swap between consecutive rows; the strands between the rows are the book’s strands.



The last row, read as loops, is the permutation we started from:



Why?

Why this proves the general fact: any permutation can be drawn as such a diagram. Nudge the strands so that no two crossings sit at the same height; then, top to bottom, the diagram is a stack of one-crossing diagrams, each an adjacent transposition, and stacking is multiplying (page 8).

Try it

Exercise 1 by hand: start from the last row above and sort the numbers back into order using only swaps of neighbours (bubble sort). Each swap is an adjacent transposition, and the list of swaps, read in reverse, is another factorisation of $(1\ 2\ 4)(3\ 5)$. Different sorting orders give different factorisations; all have an odd number of swaps.

- Each crossing in the diagram corresponds to an element of S_6 of the form $(i \ i + 1)$. Reading top to bottom (and writing right to left), we find:

$$(1 \ 2 \ 4)(3 \ 5) = (3 \ 4)(1 \ 2)(4 \ 5)(2 \ 3)(3 \ 4).$$

- The point is that multiplication in S_n can be realized by stacking diagrams. And the diagram for $(1 \ 2 \ 4)(3 \ 5)$ can be obtained by stacking the diagrams for the five permutations on the right-hand side of the equation. Since we can always make a diagram where the crossings occur at different heights, this argument can be used to show that every element of S_n is equal to a product of transpositions $(i \ i + 1)$.
- Exercise 1.** Use the idea of stacking diagrams to prove that S_n is generated by $\{(i \ i + 1) \mid 1 \leq i \leq n - 1\}$.

The picture we drew of the permutation $(1 \ 2 \ 4)(3 \ 5)$ can be called a braid diagram. See Aaron Abrams' Office Hour 18 on braid groups for more on this idea.

- Exercise 2.** We showed that S_4 is the collection of symmetries of a three-dimensional cube. Can any of the other symmetric groups be thought of as the symmetries of higher-dimensional cubes? Or other shapes?

The integers modulo n . Let n be an integer greater than 1. The *integers modulo n* is the set

$$\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n - 1\}$$

with the multiplication

$$(a, b) \mapsto \begin{cases} a + b & a + b \leq n - 1 \\ a + b - n & a + b \geq n. \end{cases}$$

The identity for $\mathbb{Z}/n\mathbb{Z}$ is 0. The inverse of 0 is 0, and the inverse of any other m is $n - m$. Associativity is a little trickier: to check that $(a + b) + c = a + (b + c)$, there are a few cases, depending on which sums are greater than n . We'll leave this as an exercise.

- Exercise 3.** Show that if m is any element of $\{0, \dots, n - 1\}$ with $\gcd(m, n) = 1$, then $\{m\}$ is a generating set for $\mathbb{Z}/n\mathbb{Z}$.

You are most familiar with $\mathbb{Z}/n\mathbb{Z}$ in three cases:

- 7**
- When $n = 12$, the multiplication we defined is clock arithmetic. For example, 3:00 plus 5:00 is 8:00, and 9:00 plus 4:00 is 1:00 (although perhaps we should think of 12:00 as 0:00 instead).
 - When $n = 10$, the multiplication we defined is an operation you use when balancing your checkbook. By just keeping track of the last digit, you have a quick first check for whether the sum of many big numbers is equal to what your bank says it is.
 - When $n = 2$, the multiplication is light switch arithmetic. Identify 1 with flipping the switch and 0 with do nothing. Then flip plus flip is the same as doing nothing, flip plus do nothing is the same as flip, etc.

1 Reading the diagram as an equation

Top crossing first, so the top crossing is the *rightmost* factor. The five crossings from top to bottom are $(3 \ 4)$, $(2 \ 3)$, $(4 \ 5)$, $(1 \ 2)$, $(3 \ 4)$; written right to left they give $(3 \ 4)(1 \ 2)(4 \ 5)(2 \ 3)(3 \ 4)$. Multiply out using the table method of page 7 to confirm it equals $(1 \ 2 \ 4)(3 \ 5)$.

2 Stacking = multiplying

Put the diagram of σ on top and the diagram of τ underneath; the combined picture is the diagram of $\tau\sigma$ (do σ , then τ). Interlude B on the next spread draws this for $(2 \ 6) \cdot (1 \ 2 \ 4)(3 \ 5)$.

3 Exercise 1: hint

This is bubble sort. Draw the permutation with straight strands and nudge the crossings so no two are at the same height; each crossing is then an adjacent transposition and the stack of them equals the permutation.

4 Exercise 2: hint

The regular tetrahedron (4 corners) has 12 rotations, and S_4 once reflections are allowed. In general the regular “simplex” with n corners has full symmetry group S_n . Higher cubes do *not* give symmetric groups: the 4-cube has 192 rotations, while $5! = 120$ and $6! = 720$.

5 $\mathbb{Z}/n\mathbb{Z}$: the definition by cases

Looking ahead

Interlude M, on the next spread, does this example slowly: what the elements really are, why there are exactly two cases, the three axioms one at a time, and a case-free proof of associativity.

The two cases just say “add, and if the result is n or more, subtract n ”. That is the remainder on division by n . Read the notation $\mathbb{Z}/n\mathbb{Z}$ aloud as “ $\mathbb{Z}$ mod n ”; the slash will be explained as a *quotient* on page 16.

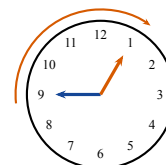
Why?

Why the fuss about associativity? Because the definition is by cases, checking $(a + b) + c = a + (b + c)$ splits into cases. The slick route: both sides equal the remainder of the integer $a + b + c$ on division by n , and taking remainders is consistent with integer addition.

6 Exercise 3: hint

$\gcd(m, n) = 1$ means there are integers u, v with $um + vn = 1$ (Bézout). Reduce mod n : $u \cdot m \equiv 1$. So m added to itself u times gives 1, and 1 generates everything. Conversely if $\gcd(m, n) = d > 1$, every multiple of m is a multiple of d mod n , so 1 is never reached.

7 Clock, cheque book, light switch



$$9 + 4 = 13 \equiv 1 \pmod{12}$$

“13” is not an element;
subtract $n = 12$ to land in $\{0, \dots, 11\}$.

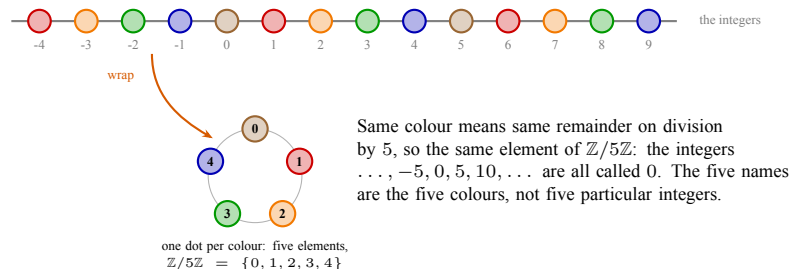
The three examples are $n = 12, 10, 2$. The $n = 2$ case, “flip plus flip is nothing”, is the group $\{0, 1\}$ and will reappear on page 15 as the target of many homomorphisms.

Interlude: M. Page 8 in detail: the integers modulo n , and why its associativity needs an argument

This is the first group on the list that is not a set of symmetries of something you can hold, and the first whose multiplication is given by a *formula with cases*. Both facts make it the example where the axioms have to be checked by hand. It is also the group that will reappear most often: as the rotations of the arrowed n -gon (page 9), as a quotient (page 16), and as the target of the sign homomorphism when $n = 2$ (page 15).

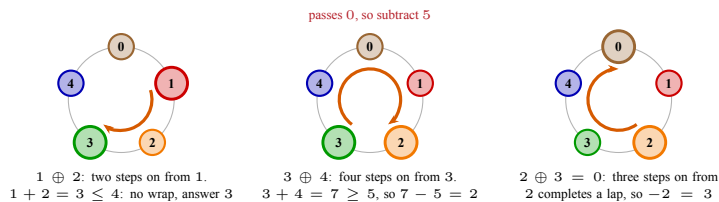
1. What the elements are

The set is $\{0, 1, \dots, n-1\}$, but each symbol is a *name for a whole family* of integers: those leaving that remainder on division by n . Take $n = 5$ and colour each integer by its remainder. Wrapping the line round a circle of five points puts one colour at each point.



2. The rule, and why there are exactly two cases

“Add, and if the answer reaches n , subtract n ”: write it $a \oplus b$. Since $a, b \leq n-1$, the ordinary sum satisfies $a+b \leq 2n-2$, so one subtraction is always enough, and the two cases of the book’s formula cover everything. The answer is exactly the remainder of $a+b$, which is why the whole thing is called *arithmetic mod n* and written $a+b \equiv c \pmod{n}$.



Watch out

The book’s word “multiplication” means the group rule, which here is *addition*. Ordinary multiplication mod n is not a group rule on this set: 0 has no multiplicative inverse. Also, $\mathbb{Z}/n\mathbb{Z}$ is not a subset of $\mathbb{Z}$ in any useful sense: $3 \oplus 4 = 2$ in $\mathbb{Z}/5\mathbb{Z}$ while $3+4=7$ in $\mathbb{Z}$.

3. The three axioms

Identity. 0 is the identity: $0 \oplus a = a \oplus 0 = a$, since $a+0 = a \leq n-1$ needs no wrap. On the circle, adding 0 is standing still.

Inverses. The inverse of 0 is 0. For $m \neq 0$ the inverse is $n-m$, because $m+(n-m) = n \geq n$, so the rule subtracts n and the answer is 0. On the circle: walking $n-m$ further from m completes one full lap. The third picture opposite is the case $m=2, n=5$.

Associativity. Here the cases bite. Both bracketings must be compared, and each $\oplus$ may or may not subtract n , so at first sight there are several combinations. A worked case, $n=5, a=3, b=4, c=2$:

$(a \oplus b) \oplus c$	$a \oplus (b \oplus c)$
$3 \oplus 4 = 7 - 5 = 2$	$4 \oplus 2 = 6 - 5 = 1$
$2 \oplus 2 = 4$ (no wrap)	$3 \oplus 1 = 4$ (no wrap)
both = 4 = remainder of $3+4+2=9$ on division by 5	

Why?

The clean argument, avoiding all cases. Each $\oplus$ either keeps the sum or subtracts n , so whichever way you bracket, the answer is the integer $a+b+c$ minus some multiple of n ; and it lies in $\{0, \dots, n-1\}$ because that is where $\oplus$ always lands. But there is only *one* integer in $\{0, \dots, n-1\}$ that differs from $a+b+c$ by a multiple of n , namely its remainder. So both bracketings equal that remainder, hence each other. The same argument in one line: $\oplus$ is “add in $\mathbb{Z}$, then take the remainder”, and taking remainders is compatible with addition. This is the substance of the exercise the book leaves you.

Big picture

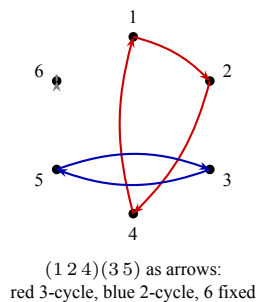
Now the extra properties. $|\mathbb{Z}/n\mathbb{Z}| = n$. It is *abelian*, since $a+b = b+a$ in $\mathbb{Z}$ and the wrap does not care about order; this makes it the simplest contrast to D_n and S_n above. It is *cyclic*: starting at 0 and adding 1 repeatedly visits every element, so $\{1\}$ generates it, and $\mathbb{Z}/n\mathbb{Z}$ is the rotation group of the arrowed n -gon on page 9. More generally m generates it exactly when $\gcd(m, n) = 1$ (Exercise 3; the picture is in Visual V2, where the subgroups turn out to be the inscribed regular polygons, one for each divisor of n). For $n=2$ it is the light switch $\{0, 1\}$, the group that records parity on page 15.

Try it

In $\mathbb{Z}/12\mathbb{Z}$: what is the inverse of 5? Of 8? Which elements are their own inverses? Check associativity on $a=7, b=9, c=8$ by both bracketings and against the remainder of 24. Then explain why $\{0, 1, \dots, n-1\}$ with ordinary subtraction-then-wrap fails the identity axiom.

Interlude: V2. Permutations as arrows; subgroups you can see

Cycle notation as a picture (page 7).

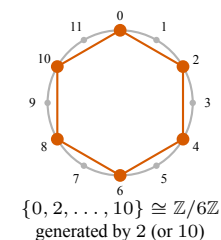
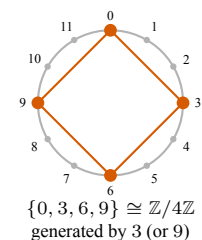
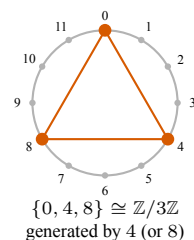
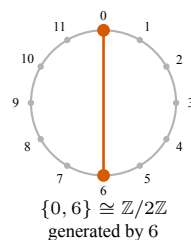


Follow the arrows: each cycle is a closed loop of arrows, and disjoint cycles are loops that share no points. The inverse permutation is the same picture with every arrow reversed: (1 4 2)(3 5).

Try it

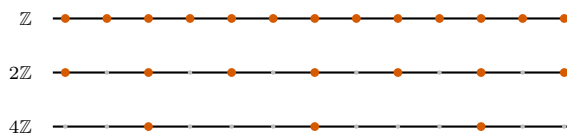
Draw (2 6) the same way, then superimpose the two pictures and trace “first the black arrows, then the (2 6) arrow” to recover (1 6 2 4)(3 5).

Subgroups of $\mathbb{Z}/12\mathbb{Z}$ (page 9). Highlighted points are closed under “add and wrap around”: each is a subgroup, and each is a regular polygon inscribed in the clock.



Exercise 3 in a picture: 5 generates all of $\mathbb{Z}/12\mathbb{Z}$ because $\gcd(5, 12) = 1$ (start at 0 and keep adding 5: 0, 5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, every point visited). Starting from 4 you only ever visit the triangle.

Subgroups of $\mathbb{Z}$ (page 9).

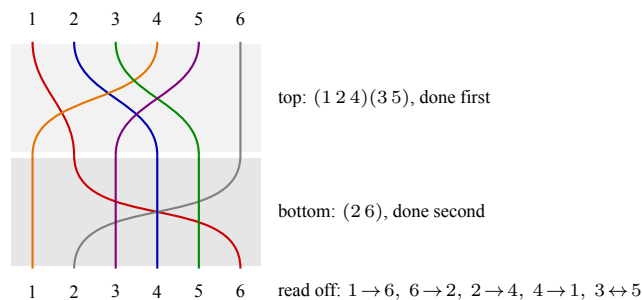


a chain of subgroups $\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z}$; each is a copy of $\mathbb{Z}$ (multiplication by 2)

Every subgroup of $\mathbb{Z}$ is $m\mathbb{Z}$ for some $m \geq 0$ (a fact worth proving: take the smallest positive element). The map $n \mapsto 2n$ is the injective homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$ of page 14, item 4: $\mathbb{Z}$ contains smaller copies of itself, something no finite group can do.

Interlude: B. Two pictures for two ideas: stacking permutation diagrams, and the Cayley tree of F_2

Stacking diagrams (page 8). The product $(2\ 6) \cdot (1\ 2\ 4)(3\ 5)$ from page 7:

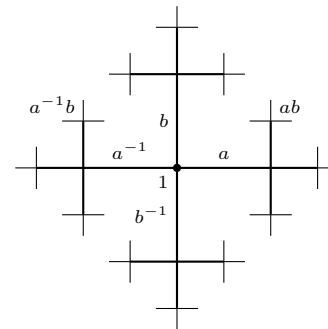


Each coloured strand can be followed from the top row to the bottom row through both blocks; the colour tells you which input it started from. The red strand starts at 1, goes to 2 in the top block, then to 6 in the bottom block: $1 \mapsto 6$. Reading all six gives $(1\ 6\ 2\ 4)(3\ 5)$, agreeing with the table on the page-7 annotations.

Why?

Why is the diagram of $\tau\sigma$ obtained by putting σ on top? Because strands are read top to bottom and σ is applied first. If you prefer to put the first-applied permutation at the bottom, every formula flips; the book's choice matches "do h , then g " = gh .

A preview of the object whose symmetry group is F_2 (page 13).

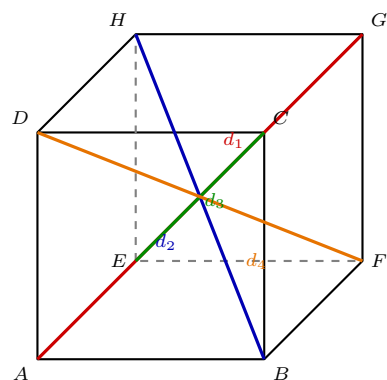


Start at the dot marked 1. Each reduced word is a path: read the letters, going right for a , up for b , left for a^{-1} , down for b^{-1} . Because words are reduced you never immediately backtrack, so distinct reduced words reach distinct dots. The picture continues forever, every dot has 4 neighbours, and there are no loops: an infinite 4-valent *tree*. This is the Cayley graph of F_2 ; Office Hour 2 shows that F_2 is exactly the group of symmetries of it that preserve the letter labels and arrow directions. Office Hours 3–6 use this tree constantly.

Looking ahead

Compare with the arrowed n -gon (page 9, symmetry group $\mathbb{Z}/n\mathbb{Z}$), the arrowed line (page 10, $\mathbb{Z}$) and the coloured grid (page 11, $\mathbb{Z}^2$). All are the same construction: one dot per group element, one coloured arrow per generator.

Interlude: B2. The dictionary: each kind of turn of the cube is a kind of permutation of the four diagonals



Corners A, B, C, D in front, E, F, G, H behind; the four long diagonals $d_1 = AG, d_2 = BH, d_3 = CE, d_4 = DF$ in their colours.

Now that cycle notation is available (page 7), the correspondence of page 4 can be written out. Each kind of turn (drawn before and after in Interludes A2–A4) produces one kind of permutation of $d_1, \dots, d_4$, and the counts match line by line.

rotation of the cube	how many	permutation of $d_1, \dots, d_4$	count
do nothing	1	identity	1
$\pm 90^\circ$ about an axis through two opposite face centres (3 axes, 2 directions)	6	4-cycle, e.g. $(d_1 d_2 d_4 d_3)$	6
180° about a face-centre axis	3	two disjoint swaps, e.g. $(d_1 d_4)(d_2 d_3)$	3
$\pm 120^\circ$ about a long diagonal (4 axes, 2 directions)	8	3-cycle, e.g. $(d_2 d_4 d_3)$; the axis diagonal is fixed	8
180° about an axis through midpoints of two opposite edges (6 axes)	6	one swap, e.g. $(d_1 d_2)$	6
total	24		24

Why?
Why does a 120° turn about d_1 fix d_1 and cycle the other three? The axis is d_1 itself, so d_1 is carried to itself. The three corners adjacent to A are B, D, E ; the turn cycles them, and d_2, d_4, d_3 are the diagonals starting at those corners. The examples in the table are the ones drawn in Interludes A2–A4, read off the before/after pictures.

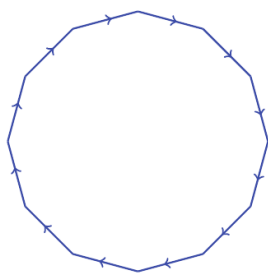
Try it
Check that the two columns of counts match line by line. The matching is not a coincidence: it is the bijection of page 4 restricted to each type of rotation. As a harder check, compose two edge-axis 180° turns about perpendicular axes and identify the result in the table.

Looking ahead
The full symmetry group of the cube (rotations and reflections, 48 elements) is not S_4 ; it is $S_4 \times \mathbb{Z}/2\mathbb{Z}$, where the extra factor records whether the map is a mirror image. Products of groups are not defined in this chapter, but the “ $\times \mathbb{Z}/2\mathbb{Z}$ ” idea is exactly the homomorphism to $\mathbb{Z}/2\mathbb{Z}$ of page 15, item 4.

- 1 The idea of light switch arithmetic inspires a whole class of groups called lamplighter groups; see Jen Taback's Office Hour 15.

For what object is $\mathbb{Z}/n\mathbb{Z}$ the group of symmetries? Using the idea of the clock we might try to use a regular n -gon. We can certainly see $\mathbb{Z}/n\mathbb{Z}$ as a set of symmetries of the n -gon, namely, the rotational symmetries. Thus, $\mathbb{Z}/n\mathbb{Z}$ can be regarded as a subgroup of the full symmetry group, D_n . This means that we have realized $\mathbb{Z}/n\mathbb{Z}$ as a subset of D_n in such a way that the multiplication in $\mathbb{Z}/n\mathbb{Z}$ agrees with the multiplication in D_n . We can see that $\mathbb{Z}/n\mathbb{Z}$ is a proper subgroup of D_n (i.e., it is not the whole thing), since D_n also has reflectional symmetries.

- 3 That's pretty good. But we can modify the n -gon so that $\mathbb{Z}/n\mathbb{Z}$ is exactly the group of symmetries. One way to do this is to draw a little arrow in the middle of each edge:



We think of the arrow as defining a preferred direction on each edge. Then we define a symmetry of this modified n -gon to be a rigid transformation that preserves the directions of the arrows. This is still a perfectly fine notion of symmetry! The group $\mathbb{Z}/n\mathbb{Z}$ is the full group of symmetries of the n -gon in this modified sense.

1.2 INFINITE GROUPS

- 4 Many undergraduate courses in abstract algebra give short shrift to infinite groups. As we will see in this book, this is where all of the fun is! Let's start with the simplest infinite group.

The integers. You are probably familiar with the set of *integers*, or whole numbers:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

- 5 The multiplication $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is the usual addition of integers:

$$(m, n) \mapsto m + n.$$

- 6 The group $\mathbb{Z}$ is generated by $\{1\}$. But it is also generated by $\{2, 3\}$, for example.

1 Lamplighter groups

Imagine a street of lamps, each on or off (each a copy of $\mathbb{Z}/2\mathbb{Z}$), and a lamplighter who walks along the street toggling lamps. The positions plus the lamp states form a group; Office Hour 15 studies it. The point here is only that "light-switch arithmetic" is a real ingredient in current research.

2 Subgroup: the definition, which the text only implies

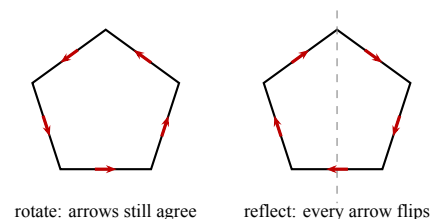
A *subgroup* of G is a subset $H \subseteq G$ that is itself a group using the same multiplication. Equivalently: $1 \in H$, and H is closed under products and inverses. "Proper" means $H \neq G$. The rotations of the n -gon are a proper subgroup of D_n because the reflections are missing.

Try it

Which of these are subgroups of $\mathbb{Z}$ (under addition)? the even integers; the odd integers; the multiples of 5; $\{-1, 0, 1\}$.

3 The arrowed n -gon

The trick is to decorate the object until exactly the symmetries you want survive. Rotations carry every arrow to an arrow pointing the same way around; a reflection reverses the sense of travel, so every arrow ends up pointing the wrong way.



So the symmetries of the decorated pentagon are precisely the 5 rotations, and those compose exactly like $\mathbb{Z}/5\mathbb{Z}$: rotate by j clicks then k clicks is $j + k$ clicks, reduced mod 5.

Big picture

This decorated polygon is the Cayley graph of $\mathbb{Z}/n\mathbb{Z}$ with generator 1: vertices are the elements $0, \dots, n-1$, and the arrow from k to $k+1$ is "add the generator". The same recipe will build an object for every group in Office Hour 2.

4 §1.2, infinite groups

Everything so far has been finite. From here on most groups are infinite, and the geometric viewpoint earns its keep: you cannot list an infinite multiplication table, but you can draw the group's picture.

5 $\mathbb{Z}$ with addition

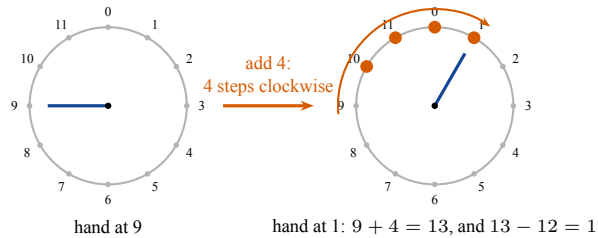
Translate the axioms: identity 0, inverse of n is $-n$, associativity is the school fact $(a + b) + c = a + (b + c)$. Because the operation is written $+$, people say "additive notation" and write ng instead of g^n .

6 Generated by $\{1\}$, and by $\{2, 3\}$

From $\{2, 3\}$: $3 - 2 = 1$, and 1 generates everything. From $\{2\}$ alone you get only the even numbers, a proper subgroup. Generating sets are far from unique; Cayley graphs depend on the generators chosen.

Interlude: E. Pages 8–9 step by step: adding on a clock, and why arrows leave only the rotations

1. Adding in $\mathbb{Z}/12\mathbb{Z}$ is moving the hand. The element a is “the hand points at a ”; adding b means moving the hand b steps clockwise. If the hand passes 12 you have gone round, and subtracting 12 just renames the hour.



Identity: add 0, the hand stays. Inverse of 4: add 8, because $4 + 8 = 12$ brings the hand back where it was. So $-4 = 8$ in $\mathbb{Z}/12\mathbb{Z}$, which is the book's $n - m$.

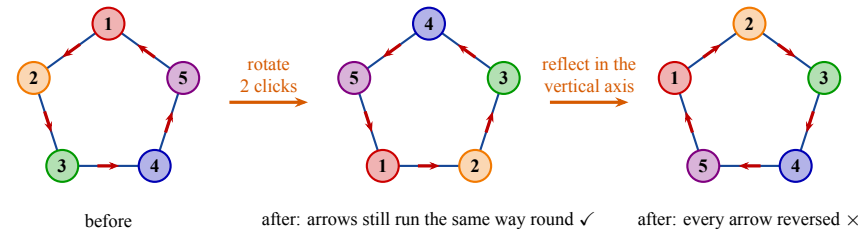
Why?

Why associativity is automatic: $(a + b) + c$ and $a + (b + c)$ are both “move the hand $a + b + c$ steps in total”, and the hour you land on depends only on the total number of steps, taken modulo 12.

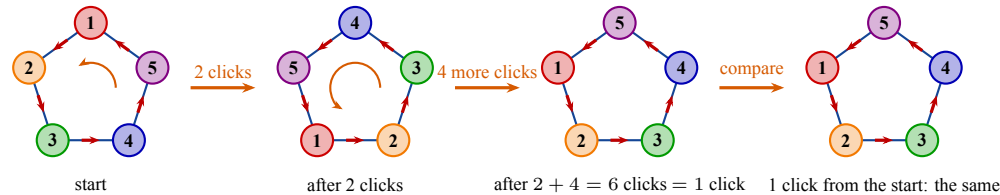
Try it

Exercise 3 on the clock: starting at 0, keep adding 5 and mark the hours you visit; then do the same with 4. With 5 every hour is marked (twelve steps, twelve different hours); with 4 only 0, 4, 8 are, because $\gcd(4, 12) = 4$ and every landing hour is a multiple of 4. Visual page V2, two spreads back, draws both.

2. The arrowed pentagon: a rotation keeps the arrows, a reflection reverses them. Corners are numbered and coloured; a symmetry moves each corner to a new position. The red arrows are part of the decorated object and must land on arrows pointing the same way round.



3. Rotations compose by adding clicks, modulo n . Two clicks then four clicks is six clicks, and six clicks of a pentagon is one click, since five clicks is a full turn: $2 + 4 = 6 \equiv 1 \pmod{5}$.



Why?

This is why the rotations of the arrowed n -gon are $\mathbb{Z}/n\mathbb{Z}$: “ k clicks” matches the element k , doing one rotation after another matches addition, and n clicks matches 0. The reflections of D_n are excluded by the arrows, so nothing else is left.

- 1 Of what object is $\mathbb{Z}$ the set of symmetries? A first guess is that $\mathbb{Z}$ is the symmetries of the number line L , by which we mean the real line $\mathbb{R}$ with a distinguished point at each integer:



Here, by a symmetry of L , we just mean a rigid transformation of L that preserves the set of distinguished points. Examples of symmetries of L are: translate L to the left or right by n units, where n is a positive integer; reflect it about an integer point; and reflect it about a half-integer point.

- 2 The group $\mathbb{Z}$ gives a set of symmetries of L as follows: the integer n is the symmetry that translates L by n units to the right if $n \geq 0$ and translates L by $-n$ units to the left if $n < 0$. If we think of L as an ∞ -gon, this is analogous to what we did for $\mathbb{Z}/n\mathbb{Z}$. As in the case of $\mathbb{Z}/n\mathbb{Z}$, this identifies $\mathbb{Z}$ as a proper subgroup of the set of symmetries of L , since we don't get any of the reflections.

- 3 We can use the same trick as in the $\mathbb{Z}/n\mathbb{Z}$ case to modify L in such a way that $\mathbb{Z}$ is the full symmetry group: on each edge we draw a little arrow that points to the right:



- 4 **The integers squared.** Our next group is

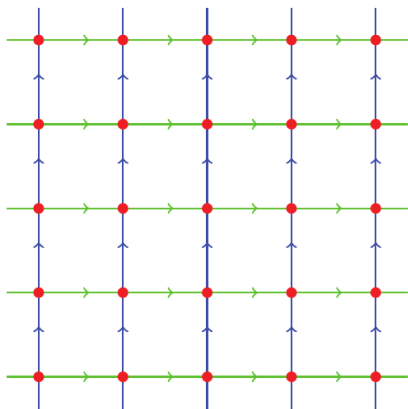
$$\mathbb{Z}^2 = \{(m, n) : m, n \in \mathbb{Z}\}$$

with the multiplication

$$((m, n), (m', n')) \mapsto (m + m', n + n').$$

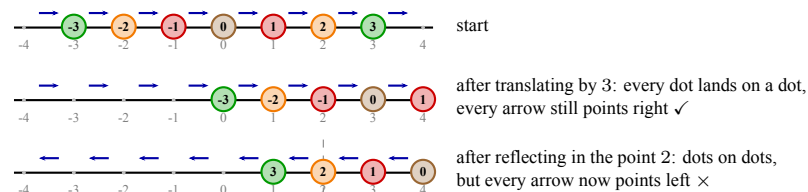
- 5 If we think of $\mathbb{Z}^2$ as the set of vectors in $\mathbb{R}^2$ with integer coordinates, then this is simply vector addition. One generating set for $\mathbb{Z}^2$ is $\{(1, 0), (0, 1)\}$.

We can use the same idea as above to build an object whose symmetries are exactly described by $\mathbb{Z}^2$:



1 The number line L and its symmetries

Rigid motions of a line are translations and reflections. Those preserving the integer points: translations by an integer, reflections in an integer point (which fixes that point), and reflections in a half-integer point (which swaps the two neighbours). The full symmetry group of L is the *infinite dihedral group* D_∞ , the $n \rightarrow \infty$ version of D_n .



Why?

Why the slide passes both tests. Translating by 3 is $x \mapsto x + 3$, the *same* shift for every point, so gaps are kept, $|(x + 3) - (y + 3)| = |x - y|$, and order is kept, $x < y \Rightarrow x + 3 < y + 3$: dots land on dots and every arrow still points right. Reflecting in 2 is $x \mapsto 4 - x$: gaps are kept, but $x < y$ becomes $4 - x > 4 - y$, so every arrow turns round.

2 $\mathbb{Z}$ as translations

Write T_n for “slide right by n ”. Then T_m followed by T_n is T_{m+n} : composition of translations matches addition of integers. In the language of page 13, $n \mapsto T_n$ is an *injective homomorphism* from $\mathbb{Z}$ into the symmetries of L , and its image is a proper subgroup (no reflections).

3 Arrows again

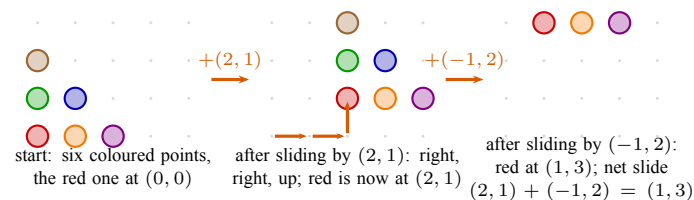
Adding the right-pointing arrows kills the reflections, exactly as on page 9. The decorated line is the Cayley graph of $\mathbb{Z}$ with generator 1.

4 $\mathbb{Z}^2$

Pairs of integers, added coordinate by coordinate: the “lattice points” of the plane. Identity $(0, 0)$; inverse of (m, n) is $(-m, -n)$. $\mathbb{Z}^2$ is abelian: $(m, n) + (m', n') = (m', n') + (m, n)$. Page 17 contrasts it with the free group F_2 , which also has two generators but is not abelian.

5 $\mathbb{Z}^2$ as slides of the plane, before and after

Each element (m, n) is the slide “ m to the right, n up”. Slides compose by adding coordinates, which is vector addition:



The generating set $\{(1, 0), (0, 1)\}$: the middle panel shows the slide $(2, 1)$ as unit steps right, right, up, so $(2, 1) = 2(1, 0) + 1(0, 1)$; every slide is such a string of unit steps, hence a sum of the two generators and their negatives. The decorated grid at the bottom of the page is the object whose symmetries are exactly these slides (page 11, Interlude F).

- 1 In order for this to work, we need to again refine what we mean by symmetry: this time we insist that colors are preserved as well as the red points and the arrows. The symmetry corresponding to $(m, n) \in \mathbb{Z}^2$ is the translation by m units to the right and n units up. (If m or n are negative, then translate left and down accordingly.)

Multiplicative groups of numbers. We would be remiss to not mention two important groups, $\mathbb{R}^\times$ and $\mathbb{C}^\times$. These are the groups of nonzero real numbers and nonzero complex numbers. The group multiplication in each group is the usual multiplication in $\mathbb{R}$ and $\mathbb{C}$. In both cases the identity is 1. An important distinction between these groups and our previous examples is that $\mathbb{R}^\times$ and $\mathbb{C}^\times$ are not finitely generated, that is, there does not exist a finite generating set.

- 2 If you've already seen these groups, you might want to learn about the quaternions, a generalization of the complex numbers which is very important in physics. Instead of just i , the quaternions have i , j , and k ! There is a corresponding multiplicative group, consisting again of the nonzero elements.

Matrix groups. There is another set of examples of groups that you learned about in linear algebra (although in such a class you might not have used the word "group"). Denote by $\text{GL}(n, \mathbb{R})$ the set of real $n \times n$ matrices with nonzero determinant. Why is this a group? Well, given two matrices, we already know how to multiply them. Note that this multiplication is closed (that is, the product of two matrices with nonzero determinant is another matrix with nonzero determinant). Indeed, this follows immediately from the following basic fact you learned in linear algebra:

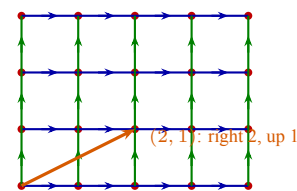
- 5
$$\det(MN) = \det(M) \det(N).$$

Two other things you learned in linear algebra is that every square matrix with nonzero determinant is invertible and that multiplication is associative. The identity matrix has determinant 1, which is not zero, so we have checked that $\text{GL}(n, \mathbb{R})$ is indeed a group. It is called the *general linear group*. There are really lots of general linear groups, since n can be any nonnegative integer and $\mathbb{R}$ can be replaced with any ring (see below), but when the context is clear we still call it *the* general linear group.

There are many variations. The group $\text{SL}(n, \mathbb{R})$ is the subgroup of $\text{GL}(n, \mathbb{R})$ consisting of matrices with determinant 1 (this is the *special linear group*). The group $\text{GL}(n, \mathbb{Z})$ is the subgroup of $\text{GL}(n, \mathbb{R})$ consisting of all invertible integral matrices (note: these are exactly the integral matrices with determinant ± 1). Then $\text{SL}(n, \mathbb{Z})$ is the subgroup of $\text{GL}(n, \mathbb{Z})$ consisting of integral matrices with determinant 1. There is also $\text{GL}(n, \mathbb{C})$, the group of complex $n \times n$ matrices with nonzero determinant, and $\text{SL}(n, \mathbb{C})$, the subgroup of $\text{GL}(n, \mathbb{C})$ consisting of $n \times n$ matrices with determinant 1. All of the above examples are subgroups of $\text{GL}(n, \mathbb{C})$, and there are many other examples of interesting subgroups of $\text{GL}(n, \mathbb{C})$ (subgroups of $\text{GL}(n, \mathbb{C})$ are called *linear groups*).

We will study $\text{SL}(2, \mathbb{Z})$ in detail in Office Hour 3, giving a finite presentation for this group as well as many of its relatives.

1 Colours and arrows on the grid



blue = horizontal edges, green = vertical edges

A rigid motion of the plane preserving the red points, the arrow directions and the colours cannot rotate (a 90° turn would swap blue and green) and cannot reflect (arrows would flip), so it is a translation, and by a lattice vector. Hence the symmetry group is exactly $\mathbb{Z}^2$.

2 $\mathbb{R}^\times$ and $\mathbb{C}^\times$ are not finitely generated

$\mathbb{R}^\times$ means the nonzero reals; zero is excluded because it has no inverse.

Why?

If finitely many numbers $r_1, \dots, r_k$ generated $\mathbb{R}^\times$, every element would be $r_1^{e_1} \dots r_k^{e_k}$ with integer exponents: only countably many possibilities, but $\mathbb{R}^\times$ is uncountable. Same for $\mathbb{C}^\times$. Even $\mathbb{Q}^\times$ is not finitely generated, since there are infinitely many primes and each needs its own generator.

3 Quaternions

Numbers $a + bi + cj + dk$ with $i^2 = j^2 = k^2 = ijk = -1$. Since $ij = k$ but $ji = -k$, the nonzero quaternions form a non-abelian group of "numbers". They encode rotations of 3-space.

4 $\text{GL}(n, \mathbb{R})$

Background you need

An $n \times n$ matrix represents a linear map of $\mathbb{R}^n$; its determinant is the factor by which the map scales volume (negative if orientation flips). For 2×2 : $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. M is invertible exactly when $\det M \neq 0$, and then $M^{-1} = \frac{1}{\det M} \text{adj}(M)$, where $\text{adj}(M)$ has integer entries whenever M does.

The page runs the standard checklist: closure (from $\det(MN) = \det M \det N$), identity, inverses, associativity (matrix product is composition of linear maps).

5 SL , and why $\text{GL}(n, \mathbb{Z})$ needs $\det = \pm 1$

If M and M^{-1} are both integer matrices, $\det M$ and $\det M^{-1}$ are integers with product 1, so $\det M = \pm 1$; conversely the adjugate formula then gives an integer inverse.

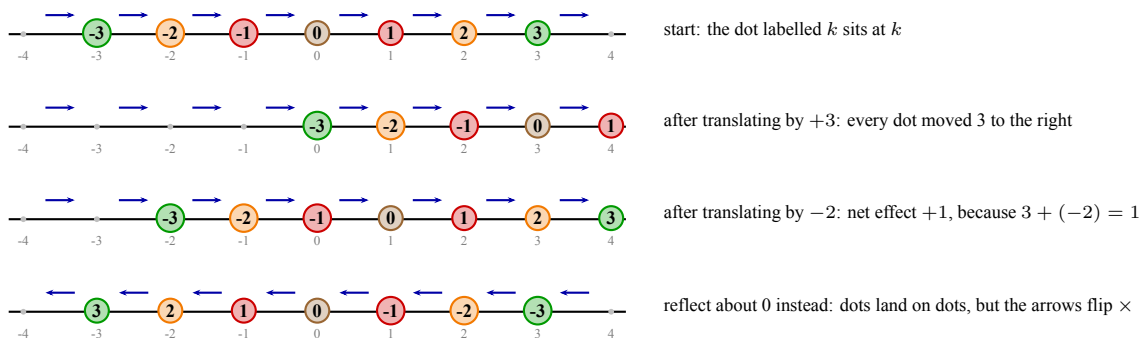
6 Linear groups

All of these sit inside $\text{GL}(n, \mathbb{C})$. Office Hour 3 studies $\text{SL}(2, \mathbb{Z})$ and finds free groups inside it: a first link between matrices and words.

Interlude: F. Pages 10–11 step by step: the integers as slides of a line, the grid as slides of a plane

Same method: label the points, colour the labels, and watch where they go. A symmetry of the marked line must send marked points to marked points; with arrows added it must also send each arrow to an arrow pointing the same way.

1. $\mathbb{Z}$ as translations of the arrowed line. The integer n acts as “slide everything n units to the right” ($-n$ units to the left if $n < 0$).



Why?

Doing slide 3 and then slide -2 is slide 1: composing translations is adding integers. That is the whole reason $\mathbb{Z}$ under addition is the symmetry group. Identity = slide 0; inverse of slide n = slide $-n$. The reflection about 0 maps marked points to marked points, so it is a symmetry of the bare line, but it reverses every arrow; adding the arrows is what cuts the symmetry group down to exactly $\mathbb{Z}$.

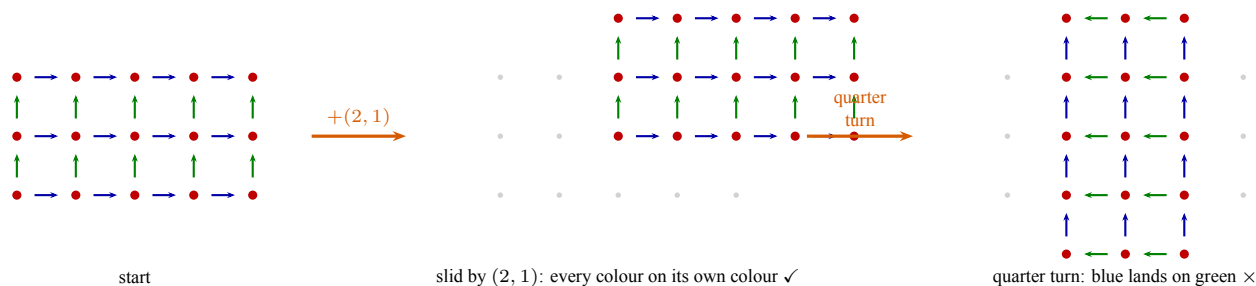
Try it

Generators: from $\{1\}$ you reach n by sliding 1 unit n times. From $\{2, 3\}$: slide 3 then slide -2 gives 1, and then everything. From $\{2\}$ alone: only even slides, a proper subgroup ($2\mathbb{Z}$).

2. $\mathbb{Z}^2$ as translations of the coloured grid. The pair (m, n) acts as “slide m to the right and n up”. Slides add coordinate by coordinate: $(2, 1)$ then $(-1, 2)$ is $(1, 3)$.

Watch out

Without colours, a quarter turn would be a symmetry of the grid (points to points, arrows to arrows). With the horizontal edges blue and the vertical ones green, the turn sends blue arrows where green ones were, so it is excluded; a reflection is excluded because it flips arrows. Only the slides survive, and they compose exactly like $\mathbb{Z}^2$.



1 **Exercise 4.** Construct a matrix group whose entries lie in a finite field.

The free group of rank 2. Our next example is the most complicated (and interesting!) one yet. And it is most probably not a group that you saw in your undergraduate algebra course.

2 First we define a *word* in the letters a and b to be an arbitrary finite string made up of the symbols a, b, a^{-1} , and b^{-1} . Some examples are:

$$abababa, aaaaaaa, aba^{-1}b^{-1}, \text{ and } aa^{-1}.$$

We allow the empty word as well; this is the string containing no letters. We can multiply two words by concatenating them:

$$(ab, b^{-1}a) \mapsto abb^{-1}a.$$

3 Next, we define a *reduced word* to be a word with the property that we never see an a followed by an a^{-1} (or vice versa) or a b followed by a b^{-1} (or vice versa). If we have a word that is not reduced, we can make it shorter by deleting an offending pair of symbols:

$$aa^{-1}abb^{-1}a \rightsquigarrow abb^{-1}a.$$

Performing this procedure inductively will eventually lead to a reduced word. You should check that this reduction procedure always results in the same reduced word. In the above example, there are three places where we could start reducing. In this case, though, no matter what we do we will end up with the reduced word aa .

Exercise 5. Show that every word in a, b, a^{-1} , and b^{-1} can be reduced to a unique reduced word.

4 We are finally ready to define the *free group of rank 2*. It is the set

$$F_2 = \{\text{reduced words in } a \text{ and } b\}$$

with the multiplication:

$$\text{concatenate, then reduce.}$$

5 Let's check that this is a group. The identity is the empty word. The inverse of a word is obtained by reversing the word, and then replacing each a by a^{-1} , a^{-1} by a , b by b^{-1} , and b^{-1} by b . For example, the inverse of

$$aba^{-1}b$$

is

$$b^{-1}ab^{-1}a^{-1}$$

6 Indeed, if we concatenate these two (in either order!) and reduce, we obtain the empty word. Associativity follows immediately from the fact that the reduction procedure always results in the same reduced word (but this fact needs to be proved!). By the very definition of F_2 , we see that this group is generated by $\{a, b\}$.

1 **Exercise 4: hint**

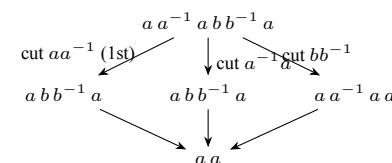
Use entries in $\mathbb{Z}/p\mathbb{Z}$ for a prime p , where every nonzero element has a multiplicative inverse. For instance $GL(2, \mathbb{Z}/2\mathbb{Z})$ consists of the invertible 2×2 matrices with entries 0, 1; there are 6 of them, and the group behaves like S_3 (it permutes the three nonzero vectors of $(\mathbb{Z}/2\mathbb{Z})^2$).

2 **Words**

A word is just a finite string over the four-letter alphabet $\{a, b, a^{-1}, b^{-1}\}$. At this stage a^{-1} is a *symbol*, not yet an inverse of anything; the notation is chosen so that it will become the inverse once the group is built. The empty word has length 0.

3 **Reduced words and reduction**

The reduction step deletes an adjacent pair xx^{-1} or $x^{-1}x$. Starting from $aa^{-1}abb^{-1}a$ there are three possible first steps, but all roads lead to aa :



That “all roads lead to the same place” is Exercise 5 and is not obvious in general; it is the whole difficulty in defining free groups.

4 **F_2 : concatenate, then reduce**

So the product of the reduced words ab and $b^{-1}a$ is: concatenate to $abb^{-1}a$, reduce to aa . The name *free* means there are no relations except the forced ones $xx^{-1} = 1$: two reduced words are equal only if they are literally the same string.

Big picture

Contrast with $\mathbb{Z}^2$, where $ab = ba$ is an extra rule. In F_2 , $ab \neq ba$ because they are different reduced words. Free groups are the “most non-commutative” groups on their generators, and §1.4 shows every group is obtained from one by imposing rules.

5 **The inverse of a word**

“Reverse the order and invert each letter” is the socks-and-shoes rule $(xy)^{-1} = y^{-1}x^{-1}$: to undo “socks then shoes” you remove shoes first. Check: $(aba^{-1}b)(b^{-1}ab^{-1}a^{-1})$ collapses from the middle outwards, bb^{-1} , then $a^{-1}a$, then bb^{-1} , then aa^{-1} , leaving the empty word.

6 **Associativity needs the uniqueness of reduction**

Looking ahead

Interlude N, next page, runs the whole check: the set, the rule, closure, identity, inverses and this associativity argument, then the properties that make F_2 special.

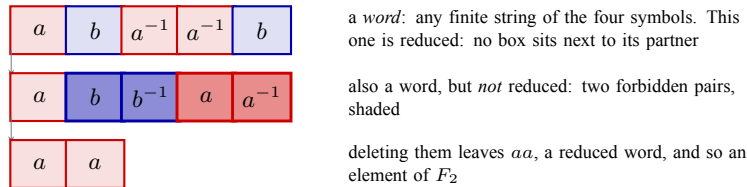
$(uv)w$ means reduce uv , append w , reduce again; $u(vw)$ reduces in a different order. Both are reductions of the single string uvw , so they agree *if* the reduced form of a string is unique. That is why Exercise 5/6 carries the weight of the construction.

Interlude: N. Page 12: the free group F_2 , first checked against the three axioms

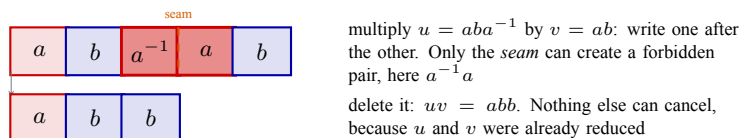
Every earlier example arrived with its elements already in hand: rotations, permutations, remainders. F_2 is the first where the *set* is defined by a condition and the *rule* is a two-step recipe, so even closure has to be argued. Each letter is drawn as a box, red for the a family and blue for the b family; shaded boxes are the pair about to be deleted.

The set, and the rule

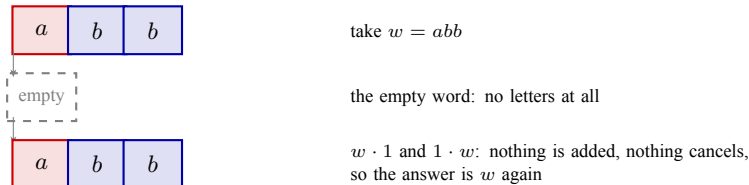
Set: the *reduced* words in a, b, a^{-1}, b^{-1} , that is, the strings in which no box stands next to its partner, together with the empty word. At this stage a^{-1} is only a *symbol*; the notation is chosen so that it will turn out to be an inverse. *Rule:* concatenate the two strings, then delete forbidden pairs until none is left.



0. Closure. Two things need checking. The deleting *stops*, since each deletion removes two boxes; and what it stops at is *reduced*, since that is exactly the condition for no deletion to be available. So the answer lies back in F_2 . Both factors were already reduced, so the only forbidden pair is at the seam, and the deleting unzips outwards from there.



1. Identity. The empty word. Writing nothing in front of or behind w changes no letter and creates no forbidden pair.



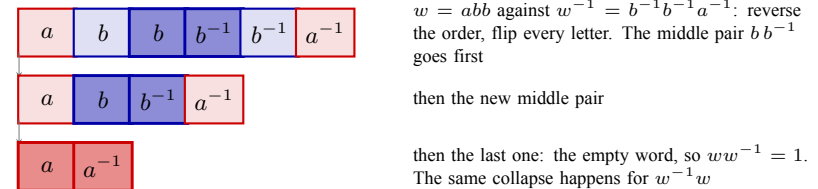
Big picture

Now the extra properties. $\{a, b\}$ generates F_2 by construction, which is what the page says next. The group is infinite: there are $4 \cdot 3^{k-1}$ reduced words of length k , since after the first letter each new letter may be anything but the partner of the one before. It is not abelian, because ab and ba are different reduced words, and no element except 1 has finite order. “Free” means precisely that the only rules are the forced ones $xx^{-1} = 1$, which is why a and b may be sent anywhere at all when you build a homomorphism out of F_2 (page 13), and why §1.4 can obtain every group by adding rules to a free one. Compare $\mathbb{Z}^2$ of page 11: same two generators, one extra rule $ab = ba$, a completely different group. Its Cayley graph is the 4-valent tree of Interlude B.

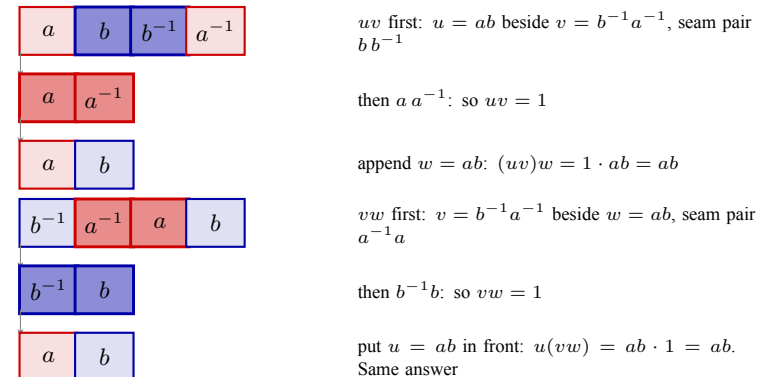
Try it

Reduce $baa^{-1}b^{-1}ba$ in two different orders and confirm both give ba . Write down the inverse of $a^{-1}bab^{-1}$ and check the collapse. How many reduced words have length 3? Finally, decide which axiom would fail if the set were taken to be *all* words rather than the reduced ones, with concatenation as the rule.

2. Inverses. Reverse the word and flip every letter: this is the socks-and-shoes rule $(xy)^{-1} = y^{-1}x^{-1}$, since to undo “socks, then shoes” you take the shoes off first. The two strings then annihilate from the middle outwards.



3. Associativity. This is the one that costs something. $(uv)w$ and $u(vw)$ are two different *orders of deleting* inside the one long string uvw , and there is no reason on the face of it that different orders give the same reduced word. Take $u = ab, v = b^{-1}a^{-1}, w = ab$:



Why?

Why it always works. Reduce the single string uvw in any order you like; the claim of Exercise 5 is that the reduced word you reach does not depend on the order. Granting that, $(uv)w$ and $u(vw)$ are both reductions of uvw , so they are the same element, which is associativity. The claim itself is Exercise 6, sketched in Interlude G: if two deletions are available at once they either sit apart, in which case doing one leaves the other untouched, or they overlap, in which case both leave the same string. Induction on length turns that into uniqueness. *This is the price of defining a group by strings rather than by functions:* for the cube, for S_n and for D_n , associativity came free from composition of functions.

- 1 **Exercise 6.** Check the details of the construction of the free group of rank 2, in particular the assertion that the multiplication is associative. *Hint: Show by induction on the length of (unreduced) words that if there are two choices for a reduction, then either choice leads to the same reduced word. There are two cases, according to whether the two reductions are adjacent or not.*

Free groups are important examples of groups in geometric group theory. Office Hours 3, 4, 5, and 6 explore these groups in more detail.

- 2 How can we construct an object whose symmetry group is F_2 ? This is not so easy! In Office Hour 2 we will explain a way to do this for an arbitrary group.
- 3 **Other free groups.** Given a set S , we can define a free group $F(S)$ as follows. First, we artificially create inverses for each element of S . For $s \in S$, we can write this inverse as s^{-1} (let's not allow s and s^{-1} to both be elements of S). Then the elements of $F(S)$ are reduced words in the elements of S , and the multiplication is defined in the same way as for F_2 . As such, when S is a finite set with n elements, we can denote $F(S)$ by F_n (different choices of S with $|S| = n$ technically give different groups, but they are all isomorphic in the sense defined below). Convince yourself that $F(S)$ still is a well-defined group when S is infinite, or even uncountable. In general, the cardinality of S is the *rank* of the corresponding free group.
- 4 As we will explain in more detail in the next section, free groups are important because every group is the quotient of a free group.
- 5 **And many more.** One aspect of geometric group theory is to explore the zoo of infinite groups that are out there. Some examples of interesting classes of groups in this book are the Coxeter groups already mentioned (Adam Piggott's Office Hour 13), the right-angled Artin groups (Bob Bell and Matt Clay's Office Hour 14), the lamplighter groups (Jen Taback's Office Hour 15), Thompson's groups (Sean Cleary's Office Hour 16), the mapping class groups (Tara Brendle and Leah Childers' Office Hour 17), and the braid groups (Aaron Abrams' Office Hour 18).

1.3 HOMOMORPHISMS AND NORMAL SUBGROUPS

- 6 We'll now take a few minutes to give a quick overview of the most basic and important ideas from a first course in group theory. If you already know all about the first isomorphism theorem and about group presentations, then you might want to skip ahead to the next office hour.

Homomorphisms. We have talked about groups and we now have some examples. Next we would like to discuss how groups relate to each other. A *homomorphism* from a group G to a group H is a function

$$f: G \rightarrow H$$

1 Exercise 6: hint, expanded

Let w be an unreduced word and suppose two different cancelling pairs are available. If the pairs do not overlap, cancel both in either order and you reach the same word, so by induction (on length) the final reduced words agree. If they overlap, the word contains $xx^{-1}x$ (or $x^{-1}xx^{-1}$); cancelling either pair leaves the same word x , so again the results agree. This “two choices can always be reconciled” argument is often called the diamond lemma.

2 An object with symmetry group F_2

Interlude B (two spreads back) already drew it: the infinite 4-valent tree. The reason it is “not so easy” is that no familiar solid or tiling has a free symmetry group; the tree has to be built from the group itself, which is what Office Hour 2 does for every group.

3 $F(S)$ and the rank

Same recipe with any alphabet S and formal inverses. Special cases: F_0 (empty alphabet) is the trivial group $\{1\}$; F_1 is $\{a^n : n \in \mathbb{Z}\}$, which page 14 identifies with $\mathbb{Z}$. “Isomorphic in the sense defined below” refers to page 14: relabelling the alphabet does not change the group in any meaningful way.

Watch out

Rank is the size of a *free* generating set. F_2 is also generated by $\{a, b, ab\}$, but that set is not free (it satisfies $ab = (a)(b)$), so it does not change the rank.

4 “Every group is the quotient of a free group”

The slogan of §1.4, proved there in three lines once quotients and homomorphisms are available (Exercise 12, page 19). Mark this sentence: it is why free groups appear in every later chapter.

5 The zoo, in one line each

Coxeter groups: generated by reflections. Right-angled Artin groups: some pairs of generators commute, others are free. Lamplighter: page 9. Thompson's groups: piecewise-linear maps of an interval, famously strange. Mapping class groups: symmetries of surfaces up to deformation. Braid groups: the diagrams of page 7 with strands allowed to pass over and under.

6 Roadmap for §1.3

The section is one chain of ideas; keep it in view while reading the next four pages:

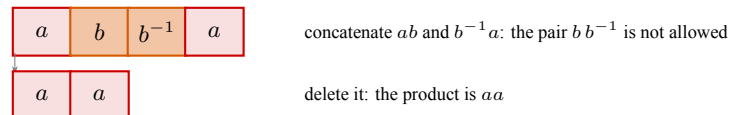
homomorphism $\rightarrow$ its kernel $\rightarrow$ kernels are normal subgroups $\rightarrow$ any normal subgroup is a kernel (build the quotient $G/N \rightarrow$ First Isomorphism Theorem).

The section starts by defining the first term of the chain.

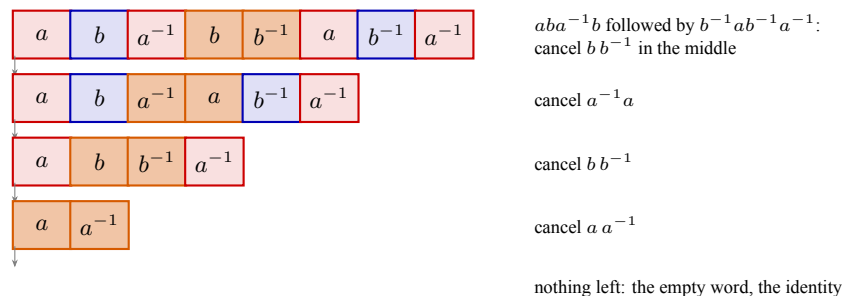
Interlude: G. Page 12 step by step: multiplying, inverting and reducing words in F_2

A word is a string of the four symbols a, b, a^{-1}, b^{-1} , drawn here as boxes (red for the a family, blue for the b family). The only rule is: a box standing next to its inverse box can be deleted. Highlighted boxes are the pair about to be deleted.

1. Multiply: concatenate, then reduce.



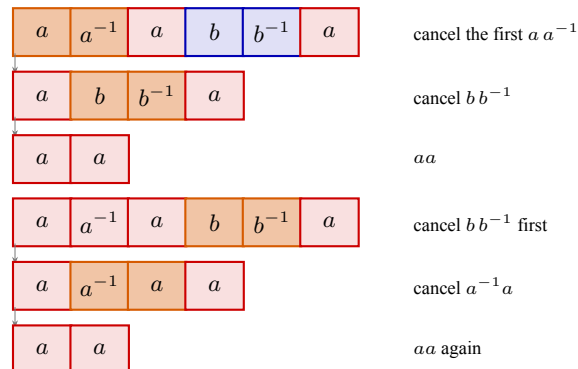
2. The inverse of a word: reverse it and invert every letter. For $aba^{-1}b$ that gives $b^{-1}ab^{-1}a^{-1}$. Check that the product really is the identity: everything cancels from the middle outwards, like taking off shoes before socks.



Watch out

Reversing without inverting, $ba^{-1}ba$, does not work: $aba^{-1}b \cdot ba^{-1}ba$ has no cancelling pair at all. Both steps are needed, and the order of the letters must be reversed because the last letter written is the first one that has to be undone.

3. Exercise 5: the reduced word does not depend on the order of cancelling. The word $aa^{-1}abb^{-1}a$ has three places to start. Two of them, side by side:



Why?

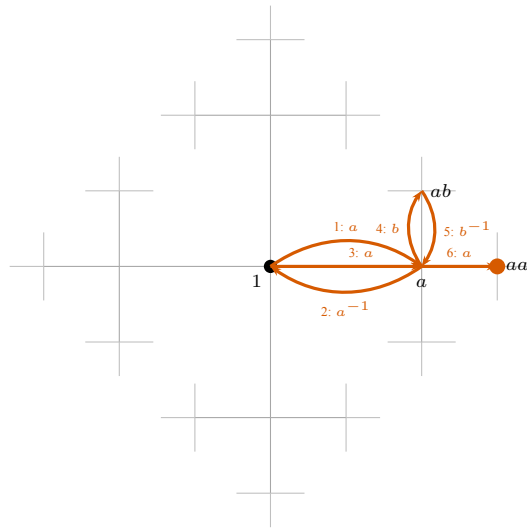
Why it always works (Exercise 6): if two cancelling pairs do not overlap, deleting one leaves the other in place, so you can delete both in either order and reach the same word. If they overlap, the word contains $xx^{-1}x$ (or $x^{-1}xx^{-1}$), and deleting either pair leaves the single letter x : the same result again. Induction on the length of the word finishes the argument, and this uniqueness is exactly what makes the multiplication associative: $(uv)w$ and $u(vw)$ are two ways of reducing the one string uvw .

Try it

Reduce $baa^{-1}b^{-1}ba$ in two different orders and check you get ba both times. Then compute $(ba)^{-1}$ and multiply to confirm you get the empty word.

Interlude: V3. Words as paths: reduction, inverses, and why $ab \neq ba$ in F_2

A word is a walk in the tree (page 12). Read $a a^{-1} a b b^{-1} a$ letter by letter, starting at 1: right for a , left for a^{-1} , up for b , down for b^{-1} .

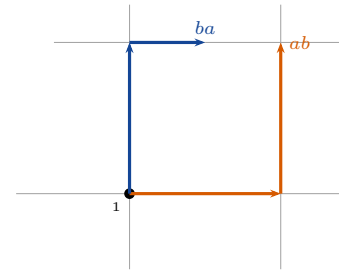


Steps 1–2 go out and straight back; so do steps 4–5. Deleting a cancelling pair xx^{-1} is exactly erasing a there-and-back. What remains is the straight path aa to the same endpoint. In a tree there is only one path without backtracking between two vertices, so the reduced word reached from any unreduced word is unique: that is Exercise 5, seen geometrically.

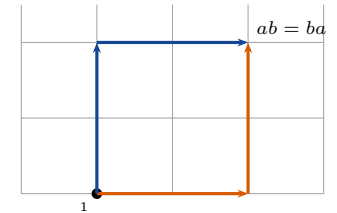
Why?

Why is the inverse of a word its reversal with letters inverted (page 12)? Because undoing a walk means retracing it backwards, and retracing a step labelled x is a step labelled x^{-1} .

Free versus abelian (pages 12, 17, 19).



in the tree of F_2 :
 ab and ba end at different vertices



in the grid of $\mathbb{Z}^2$:
the relator $aba^{-1}b^{-1}$ is a closed square

In the tree, walking a then b and walking b then a end at different places, so $ab \neq ba$ in F_2 . Imposing the relator $aba^{-1}b^{-1}$ (page 18) means declaring the walk a, b, a^{-1}, b^{-1} to be a closed loop. Gluing the tree along all such loops folds it into the square grid, the picture of $\mathbb{Z}^2$ from page 11. This is what “quotient of a free group” (page 19) looks like: the tree is the free group, the loops are the relators, and the folded picture is the group presented.

Looking ahead

Every presentation $\langle S \mid R \rangle$ has such a picture: start with the tree of $F(S)$ and fold along the loops read off from R . The result is the Cayley graph of the group (Office Hour 2). Visual page V5 shows several.

so that

$$f(ab) = f(a)f(b)$$

for all $a, b \in G$. We can say this succinctly as: f is a function that preserves the multiplication in G . A homomorphism is an *isomorphism* if it is bijective, and two groups are *isomorphic* if there is an isomorphism from one to the other. Convince yourself that isomorphic groups are essentially the same group.

To illustrate the idea of an isomorphism, let's look at a pair of groups. Consider the group $\mathbb{Z}$ on one hand, and the free group F_1 on one generator a on the other hand. Elements of F_1 are finite, freely reduced strings in a and a^{-1} , and the multiplication is given as usual by concatenation and free reduction. What is an isomorphism $\mathbb{Z} \rightarrow F_1$? (There is more than one!)

Besides isomorphisms, we might also be interested in injective homomorphisms and surjective homomorphisms. A homomorphism is injective if and only if the preimage of the identity in H is the identity in G . (Why?)

Note that any homomorphism $f: G \rightarrow H$ can be turned into a surjective homomorphism by replacing H with $f(G)$. For that reason, the surjectivity of a homomorphism is not nearly as crucial as the injectivity. As such we will focus on injective and non-injective homomorphisms.

Injective homomorphisms. If we have an injective homomorphism $f: G \rightarrow H$, then we can think of f as realizing G as a subgroup of H . Here are a few examples:

1. There is an injective homomorphism $\mathbb{Z}/n\mathbb{Z} \rightarrow D_n$ where m maps to rotation by $2\pi m/n$.
2. There is an injective homomorphism $\mathbb{Z}/2\mathbb{Z} \rightarrow D_n$ where the nontrivial element of $\mathbb{Z}/2\mathbb{Z}$ maps to any reflection.
3. There is an injective homomorphism $\mathbb{Z}/2\mathbb{Z} \rightarrow S_n$ where the nontrivial element of $\mathbb{Z}/2\mathbb{Z}$ maps to any product of disjoint transpositions.
4. For any integer $m \neq 0$, there is an injective homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$ given by multiplication by m .
5. For any choice of $(a, b) \neq (0, 0)$, there is an injective homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}^2$, where n maps to (na, nb) .
6. For any choice of nontrivial element $w \in F_2$, there is an injective homomorphism $\mathbb{Z} \rightarrow F_2$ given by $n \mapsto w^n$.

Exercise 7. How many other injective homomorphisms can you find between the groups we have already discussed?

Non-injective homomorphisms. Non-injective homomorphisms can be just as interesting and just as useful. Since we have multiple elements of G mapping to the same element of H , we are definitely losing some information. But the fact that a homomorphism preserves the multiplication of G means we are still remembering some important data. Often we can set up a non-injective homomorphism to

1 Homomorphism: " $f(ab) = f(a)f(b)$ "

Multiply first and then apply f , or apply f to each and then multiply: same answer.

$$\begin{array}{ccc} (a, b) \in G \times G & \xrightarrow{f \text{ on each}} & (f(a), f(b)) \in H \times H \\ \text{multiply in } G \downarrow & & \downarrow \text{multiply in } H \\ ab \in G & \xrightarrow{f} & f(a)f(b) = f(ab) \in H \end{array}$$

Two consequences worth proving once: $f(1) = 1$ (because $f(1) = f(1 \cdot 1) = f(1)f(1)$, then cancel), and $f(g^{-1}) = f(g)^{-1}$ (because $f(g)f(g^{-1}) = f(gg^{-1}) = f(1) = 1$). So a homomorphism automatically respects identity and inverses.

2 Isomorphism: the same group with new names

$\mathbb{Z} \rightarrow F_1$: send n to a^n (with a^{-3} meaning $a^{-1}a^{-1}a^{-1}$). Addition of exponents is concatenation of strings, so $f(m+n) = a^{m+n} = a^m a^n$. It is a bijection. The second isomorphism is $n \mapsto a^{-n}$; there are no others (an isomorphism must send the generator 1 to a generator of F_1 , and only a, a^{-1} generate).

3 "Injective iff the preimage of 1 is $\{1\}$ "

Why?

If $f(g) = f(h)$ then $f(gh^{-1}) = f(g)f(h)^{-1} = 1$, so gh^{-1} lies in the preimage of 1. If that preimage is just $\{1\}$, then $gh^{-1} = 1$, i.e. $g = h$: injective. The converse is immediate. This preimage is the *kernel*, named on page 15; so the slogan is "injective iff trivial kernel", the same test as "null space is zero" for linear maps.

4 Surjectivity is cheap

The image $f(G)$ is always a subgroup of H (products and inverses of images are images, by the two consequences in 1). Shrinking the target to $f(G)$ makes f onto without changing anything essential. That is why the book sorts homomorphisms only into injective and non-injective.

5 The six injective examples

(1) rotation by $2\pi m/n$ has order n , matching $\mathbb{Z}/n\mathbb{Z}$. (2), (3): a reflection, or a product of disjoint swaps, squares to 1, so it is a copy of $\mathbb{Z}/2\mathbb{Z}$. (4) multiplication by $m \neq 0$ is injective on $\mathbb{Z}$; its image $m\mathbb{Z}$ is a proper subgroup unless $m = \pm 1$, so $\mathbb{Z}$ contains copies of itself. (5) the line through (a, b) inside the lattice. (6) needs $w^n \neq 1$ for $n \neq 0$: true because a nontrivial reduced word, after cyclically trimming any $x \dots x^{-1}$ ends, has powers that never fully cancel.

Try it

Exercise 7 starter: find injective homomorphisms $D_3 \rightarrow S_3$, $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$, $F_2 \rightarrow F_3$, and $\mathbb{Z}^2 \rightarrow \mathbb{Z}^3$.

6 Non-injective homomorphisms

The theme of the next page: a homomorphism that identifies elements is a deliberate *forgetting* of information, and the homomorphism property guarantees the information kept is consistent with multiplication.

remember exactly the data we care about and throw away everything else. Here are some examples:

1. The most basic example is the homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ where 1 maps to 1. A light switch is a great example of this: it does not remember how many times it has been flipped, just whether it has been flipped an even or an odd number of times.
2. There is a homomorphism $\mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ given by the determinant. So if you want to know the determinant of a product of matrices, you do not need to remember the actual matrices, just their determinants.
3. There is a homomorphism $\mathbb{R}^* \rightarrow \mathbb{Z}/2\mathbb{Z}$ where positive numbers map to 0 and negative numbers map to 1. The fact that this is a homomorphism can be rephrased as: in order to tell if a product of nonzero numbers is positive or negative, you do not need to remember what the numbers are, just whether they are positive or negative.
4. The composition $\mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^* \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a homomorphism that remembers whether an element of $\mathrm{GL}(n, \mathbb{R})$ has positive or negative determinant.
5. There is a homomorphism $S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ that records whether the diagram for a permutation (as above) has an even or an odd number of crossings. In other words, this homomorphism records the parity of the number of transpositions needed to write a given element of S_n .
6. There is a homomorphism $D_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ where rotations map to 0 and reflections map to 1. This homomorphism remembers whether the polygon has been flipped over or not.
7. There is a homomorphism $F_2 \rightarrow \mathbb{Z}^2$ where the generators a and b for F_2 map to $(1, 0)$ and $(0, 1)$, respectively. This homomorphism remembers the sum of the exponents on a and b in any element of F_2 .
8. Any linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is a homomorphism. For example, the map $\mathbb{R}^2 \rightarrow \mathbb{R}$ that projects to the x -axis is a homomorphism that remembers just the first coordinate. This is a fancy way to state the obvious fact that in order to know the first coordinate of a sum of vectors, you need to know just the first coordinate of each vector.

Exercise 8. How many other non-injective homomorphisms can you find between the groups we have already discussed?

Normal subgroups. Next we will present a definition that will seem unrelated to homomorphisms; and then we will make a deep connection. A *normal subgroup* of a group G is a group N where N is a subset of G and where gng^{-1} lies in N for all $n \in N$ and all $g \in G$.

Why are normal subgroups important? Well, one way to make a normal subgroup of G is to find a group homomorphism f from G to another group H . The *kernel* of f is the set of $g \in G$ so that $f(g)$ is the identity (we use the same terminology in linear algebra). We have the following basic fact: the kernel of a homomorphism $f : G \rightarrow H$ is a normal subgroup of G .

1 The eight “forgetful” maps

Each example says: to know property P of a product, it is enough to know P of the factors. That sentence is exactly $f(ab) = f(a)f(b)$ with $f = P$. (1) parity of an integer; (2) determinant; (3) sign of a real number; (4) sign of the determinant, a composition of (2) and (3), and a composition of homomorphisms is a homomorphism; (6) has the polygon been flipped; (8) any linear map (linearity $f(u + v) = f(u) + f(v)$ is the homomorphism property in additive notation).

2 Item 5: the sign of a permutation

The map records whether the braid diagram has an even or odd number of crossings.

Watch out

There is a hidden claim: the same permutation can be drawn with different numbers of crossings (add two crossings that undo each other), so why is the *parity* always the same? Standard proofs count inversions, pairs $i < j$ with $\sigma(i) > \sigma(j)$; each adjacent swap changes the inversion count by exactly ± 1 . The book is skipping this; it is a good exercise.

3 Item 7: exponent sums

$a \mapsto (1, 0), b \mapsto (0, 1)$ extends to the map that sends a word to (total exponent of a , total exponent of b). E.g. $ab^{-1}a^2b \mapsto (3, 0)$. Two different words with the same exponent sums are identified; that is precisely the “forgetting” of the order of letters, i.e. forgetting non-commutativity. This map is called the *abelianisation* of F_2 .

4 Normal subgroup

The condition $gng^{-1} \in N$ for all g says: N looks the same from every element’s point of view. The element gng^{-1} is the *conjugate* of n by g ; think of it as “ n , performed in the coordinates that g sets up” (move by g^{-1} , do n , move back by g).

Why?

In an abelian group $gng^{-1} = n$, so every subgroup is normal. A subgroup that is not normal: in D_3 (triangle), $N = \{1, s\}$ for a reflection s . Conjugating s by a rotation r gives the reflection in the rotated axis, which is not s and not 1. So “normal” is a real restriction.

5 Kernel

The set of elements that f sends to the identity; “same terminology as linear algebra” because for a linear map it is the null space. The kernel measures how much f forgets: trivial kernel means nothing is forgotten (injective, 3 on page 14). The basic fact stated here is proved at the top of page 16.

Try it

Before turning the page, guess the kernel of each of the eight maps. Then check against the list on page 16.

Interlude: H. Pages 14–15 step by step: what “ $f(ab) = f(a)f(b)$ ” looks like on actual numbers

A homomorphism is a map you may apply either before or after multiplying and get the same answer. The only way to feel this is to test it on inputs. Three of the book’s examples, tested.

1. Parity, $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$. Send even numbers to 0 and odd numbers to 1. Add first and then take parity, or take parities first and then add in $\mathbb{Z}/2\mathbb{Z}$:

in $\mathbb{Z}$		in $\mathbb{Z}/2\mathbb{Z}$
$7 + 5 = 12$	$\xrightarrow{\text{parity}}$	$1 + 1 = 0 \checkmark$
$7 \cdot (-3) = -21$	$\xrightarrow{\text{parity}}$	$1 \cdot 1 = 1 \checkmark$
$4 + 6 = 10$	$\xrightarrow{\text{parity}}$	$0 + 0 = 0 \checkmark$

The map forgets everything about a number except whether it is even, and that one bit is enough to predict the parity of a sum. Kernel: the even numbers $2\mathbb{Z}$ (everything sent to 0).

2. Determinant, $\text{GL}(2, \mathbb{R}) \rightarrow \mathbb{R}^\times$. Take $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ and $N = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$. Then $\det M = 2$, $\det N = 2$, and $MN = \begin{pmatrix} 5 & 2 \\ 3 & 2 \end{pmatrix}$ has determinant $10 - 6 = 4 = 2 \cdot 2 \checkmark$. The determinant forgets the matrix and keeps only the area-scaling factor, and factors multiply. Kernel: the matrices of determinant 1, which is $\text{SL}(2, \mathbb{R})$.

Why?

Two consequences that hold for every homomorphism: $f(1) = 1$, because $f(1) = f(1 \cdot 1) = f(1)f(1)$ and you can cancel one $f(1)$; and $f(g^{-1}) = f(g)^{-1}$, because $f(g)f(g^{-1}) = f(gg^{-1}) = f(1) = 1$. Check them on parity: $f(0) = 0$ and $f(-7) = 1 = -f(7)$ in $\mathbb{Z}/2\mathbb{Z}$.

3. Sign of a permutation, $S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$. Count the swaps needed to build a permutation; send it to 0 if the count is even, 1 if odd. Test on the product from page 7:

(2 6): one swap, so **odd**

(1 2 4)(3 5) = (3 4)(1 2)(4 5)(2 3)(3 4): five swaps, so **odd**

product (1 6 2 4)(3 5): a 4-cycle is 3 swaps, a 2-cycle is 1 swap, total 4, so **even**

odd · odd = even, i.e. $1 + 1 = 0$ in $\mathbb{Z}/2\mathbb{Z}$: the sign respects multiplication ✓

Kernel: the permutations that need an even number of swaps, the alternating group A_n ; A_6 has $6!/2 = 360$ elements.

Watch out

Hidden claim: the same permutation can be written with different numbers of swaps (add a swap and undo it), so why is the *parity* of the count always the same? Count inversions instead: pairs $i < j$ with $\sigma(i) > \sigma(j)$. Each adjacent swap changes the number of inversions by exactly ± 1 , so its parity is fixed by σ alone. For (1 2 4)(3 5) the inversions of 2, 4, 5, 1, 3, 6 are (2, 1), (4, 1), (4, 3), (5, 1), (5, 3): five, odd, matching the five crossings.

Try it

Injective examples as pictures: $\mathbb{Z}/5\mathbb{Z} \rightarrow D_5$, $k \mapsto$ “ k clicks”, is the top row of Interlude D (five rotations, all different). $\mathbb{Z} \rightarrow \mathbb{Z}^2$, $n \mapsto (2n, n)$, marks a line of lattice points on the grid of Interlude F. Check for each that only 0 maps to the identity.

- 1 Here is why: if n is in the kernel of f , then

$$\begin{aligned} f(gng^{-1}) &= f(g)f(n)f(g^{-1}) = f(g)1f(g^{-1}) \\ &= f(g)f(g^{-1}) = f(gg^{-1}) = f(1) = 1, \end{aligned}$$

and so gng^{-1} is also in the kernel of f . Thus the kernel of f is normal.

The above examples of non-injective homomorphisms have kernels as follows:

- 2
1. The kernel of our $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is $2\mathbb{Z}$.
 2. The kernel of our $\mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ is $\mathrm{SL}(n, \mathbb{R})$.
 3. The kernel of our $\mathbb{R}^* \rightarrow \mathbb{Z}/2\mathbb{Z}$ is $\mathbb{R}_+$.
 4. The kernel of our $\mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^* \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the subgroup $\mathrm{GL}^+(n, \mathbb{R})$ consisting of matrices with positive determinant.
 5. The kernel of our $S_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the *alternating group* A_n ; this consists of the elements that can be written as the product of an even number of transpositions.
 6. The kernel of our $D_n \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the subgroup consisting of rotations; this subgroup is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.
- 3
7. The kernel of our $F_2 \rightarrow \mathbb{Z}^2$ is the set of elements whose a -exponents and b -exponents both add to zero, e.g., $a^7b^{-5}a^{-10}b^5a^3$. This group is also known as the commutator subgroup of F_2 and is denoted by $[F_2, F_2]$.
 8. The kernel of the projection $\mathbb{R}^2 \rightarrow \mathbb{R}$ is the same as the kernel from linear algebra; in this case it is isomorphic to $\mathbb{R}$.

So non-injective homomorphisms give us nontrivial normal subgroups. Can we go the other way? That is, if N is a normal subgroup of G , is there a homomorphism $G \rightarrow H$ so that N is the kernel? The answer is yes! Let's explain this. The first

- 4 step is to define the *quotient group* G/N . Informally, this is the group obtained from G by declaring every element of N to be trivial.

Let us make this more precise. Declare two elements g_1, g_2 of G to be equivalent if $g_1g_2^{-1} \in N$, and write $[g]$ for the equivalence class of g . Then the elements of G/N are the equivalence classes of the elements of G (notice that the equivalence class of the identity is precisely N). Now we need to say how to multiply two equivalence classes. We declare that the product of the equivalence class $[g_1]$ with the equivalence class $[g_2]$ is the equivalence class $[g_1g_2]$. This is precisely the place where the definition of a normal subgroup comes from—it is exactly what is needed so that this multiplication is well defined.

Exercise 9. Check that the multiplication in G/N is well defined if and only if N is normal.

There is an obvious homomorphism $G \rightarrow G/N$, where $g \in G$ maps to $[g] \in G/N$. It is not hard to see that the kernel of this homomorphism is precisely N . So

- 6 we have completed the loop: normal subgroups and surjective homomorphisms are the same thing!

The first isomorphism theorem. The so-called first isomorphism theorem succinctly summarizes the relationship between homomorphisms and normal subgroups that we have just described.

1 The proof that kernels are normal, step by step

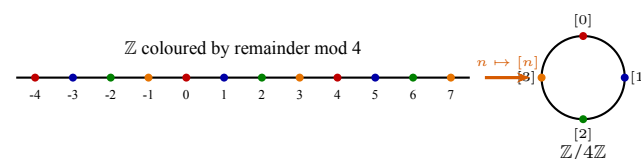
$f(gng^{-1})$ splits as $f(g)f(n)f(g^{-1})$ by the homomorphism property used twice; $f(n) = 1$ because n is in the kernel; $f(g)f(g^{-1})$ recombines to $f(gg^{-1}) = f(1) = 1$ (the two consequences from page 14). So gng^{-1} is also sent to 1.

2 The eight kernels

(1) even integers $2\mathbb{Z}$. (2) $\mathrm{SL}(n, \mathbb{R})$, so the special linear group is the kernel of $\det$. (5) the alternating group A_n , with $n!/2$ elements. (6) rotations. (7) words with both exponent sums zero, e.g. $aba^{-1}b^{-1}$: these are exactly the products of *commutators* $[x, y] = xyx^{-1}y^{-1}$, hence the name commutator subgroup. (8) the y -axis, a copy of $\mathbb{R}$.

3 Quotient group G/N : read it as “ G with N squashed to a point”

The relation $g_1 \sim g_2 \iff g_1g_2^{-1} \in N$ is an equivalence relation (reflexive since $1 \in N$; symmetric since N has inverses; transitive since N is closed under products). The classes are the *cosets* $Ng = \{ng : n \in N\}$. For $\mathbb{Z}$ and $N = 4\mathbb{Z}$:



The line wraps around the circle; each colour is one coset, and $\mathbb{Z}/4\mathbb{Z}$ is the set of four colours.

4 Why normality is exactly what is needed

Try to multiply classes by multiplying representatives. Replace g_1 by another representative n_1g_1 and g_2 by n_2g_2 . Their product is $n_1g_1n_2g_2 = n_1(g_1n_2g_1^{-1})g_1g_2$. This lies in the class of g_1g_2 exactly when $g_1n_2g_1^{-1} \in N$, which is the normality condition. That is Exercise 9 in one line; the converse direction (well defined $\implies$ normal) takes $g_1 = g, g_2 = 1, n_2 = n$.

5 Exercise 9

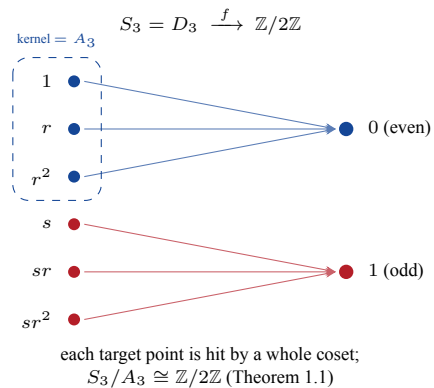
Write out the computation of 4 carefully in both directions. This is the single most important calculation in the section.

6 “Completed the loop”

Kernels are normal subgroups (page 15/16), and every normal subgroup N is the kernel of $G \rightarrow G/N$. So the two notions are interchangeable: whenever you meet a normal subgroup, there is a surjective homomorphism hiding behind it, and vice versa. The next theorem states this as an isomorphism.

Interlude: V4. Homomorphisms, kernels and cosets, element by element

A non-injective homomorphism, drawn (pages 15–17). Take $D_3 (= S_3)$ and the “flipped or not” map to $\mathbb{Z}/2\mathbb{Z}$.

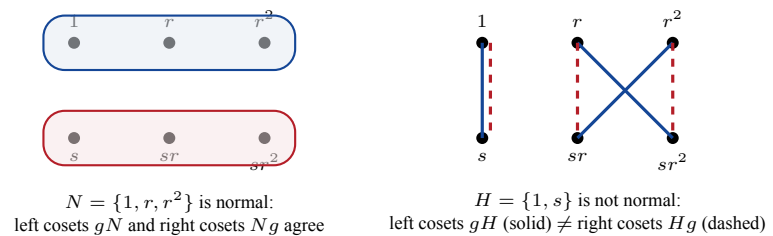


The kernel is the set of arrows landing on 0. The other three elements all land on 1: they form the coset sA_3 . A homomorphism “forgets” by merging each coset into a single point, and the merged points multiply like the quotient group. Read the coloured blocks of the D_3 table on page V1 again: they are these two cosets.

Try it

Draw the same picture for $\det : \text{GL}(2, \mathbb{R}) \rightarrow \mathbb{R}^\times$ schematically: the kernel $\text{SL}(2, \mathbb{R})$ is the fibre over 1, and the fibre over c is the coset of all matrices with determinant c .

Normal versus not normal (page 15). Two subgroups of D_3 and their cosets.



For $N = \{1, r, r^2\}$ the left and right cosets coincide, so “multiply cosets by multiplying representatives” is unambiguous and D_3/N is a group with two elements. For $H = \{1, s\}$ the partitions disagree: $rH = \{r, sr^2\}$ but $Hr = \{r, sr\}$. Try to multiply the “coset of r ” by the “coset of s ”: using representatives $r \cdot s = sr^2$, but using $sr^2 \cdot s = r$, two different answers in two different cosets. That is exactly the failure that Exercise 9 (page 16) and the normality condition $gng^{-1} \in N$ rule out.

Why?

Why is $H = \{1, s\}$ not normal, by the definition? Conjugate s by r : $rsr^{-1} = rsr^2 = sr^2 \cdot r^2 = sr$, a different reflection, not in H .

1 THEOREM 1.1 (First isomorphism theorem). If $G \rightarrow H$ is a surjective homomorphism with kernel K , then H is isomorphic to G/K .

Again using our above examples, we have the following isomorphisms:

- 2 1. $(\mathbb{Z})/(2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ (this explains the notation $\mathbb{Z}/2\mathbb{Z}$)
- 2 2. $\mathrm{GL}(n, \mathbb{R})/\mathrm{SL}(n, \mathbb{R}) \cong \mathbb{R}^*$
- 3 3. $\mathbb{R}^*/\mathbb{R}_+ \cong \mathbb{Z}/2\mathbb{Z}$
- 4 4. $\mathrm{GL}(n, \mathbb{R})/\mathrm{GL}^+(n, \mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$
- 5 5. $S_n/A_n \cong \mathbb{Z}/2\mathbb{Z}$
- 6 6. $D_n/(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$
- 7 7. $F_2/[F_2, F_2] \cong \mathbb{Z}^2$
- 8 8. $\mathbb{R}^2/\mathbb{R} \cong \mathbb{R}$

Exercise 10. Use the first isomorphism theorem to come up with your own isomorphisms.

1.4 GROUP PRESENTATIONS

You may not have covered group presentations in your first course in abstract algebra, but they are closely tied to the other topics in this section and will loom large in this book.

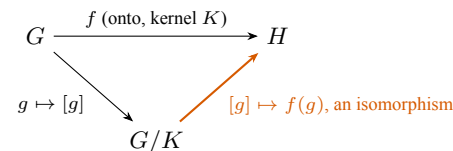
- 3 One naive way to completely describe a group is to list all of the elements and write down the multiplication table (that is, explicitly give the product of any two elements). For a very small finite group, this is a reasonable way of understanding the group. But once you start considering larger—or even infinite—groups this approach becomes unwieldy.

We already have a more economical way of listing the elements of a group: we can list a set of generators. Often the infinite groups we care about have finite sets of generators. This is good, but it certainly doesn't tell us enough about a group: the free group F_2 and the free abelian group $\mathbb{Z}^2$ both have generating sets with two elements and we know that these are different groups since one is abelian and the other is not. We need a way of describing the multiplication table. But obviously we don't want to just list all of the entries.

- 5 The key observation is that some entries in the multiplication table imply other entries. In other words, equations between group elements imply other equations. For instance, if $gh = k$ and $h = pq$, then (substituting pq for h in the first equation) we know that $gpq = k$. So what we can hope for is that we don't need to write down the entire multiplication table, but rather a small set of equalities that imply all the others. So now we have exactly hit upon the idea of what a group presentation is: it is firstly a list of generators and secondly a list of defining equalities so that all equalities follow from the listed ones.

- 6 Here is a first example, the group $\mathbb{Z}/n\mathbb{Z}$. Let's denote the elements of $\mathbb{Z}/n\mathbb{Z}$ by $a^0, \dots, a^{n-1}$ instead of $0, \dots, n-1$. Naturally, the multiplication is $a^j a^k = a^{j+k}$, where the exponents are all taken modulo n . With this notation, $\mathbb{Z}/n\mathbb{Z}$ has the

1 Theorem 1.1, with a picture and a proof sketch



Define $\phi : G/K \rightarrow H$ by $\phi([g]) = f(g)$. Well defined: if $[g] = [g']$ then $g' = kg$ with $k \in K$, so $f(g') = f(k)f(g) = f(g)$. Homomorphism: $\phi([g][g']) = \phi([gg']) = f(gg') = f(g)f(g')$. Onto: because f is. One-to-one: $\phi([g]) = 1$ means $f(g) = 1$, i.e. $g \in K$, i.e. $[g] = [1]$; now use the injectivity test of page 14. So ϕ is an isomorphism.

2 The eight isomorphisms, read as slogans

(1) integers modulo evens = parity. (2) matrices modulo $\det = 1$ = the determinant. (3) reals modulo positives = sign. (5) permutations modulo even ones = even/odd. (6) D_n modulo rotations = flipped or not. (7) F_2 modulo commutators = $\mathbb{Z}^2$: forcing $ab = ba$ in F_2 produces $\mathbb{Z}^2$, which is exactly the presentation story of the next section. (8) plane modulo a line = a line.

Try it

Exercise 10 starters: $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ by parity has kernel $\{0, 2, 4\}$, so $(\mathbb{Z}/6\mathbb{Z})/\{0, 2, 4\} \cong \mathbb{Z}/2\mathbb{Z}$. What do you get from $\mathbb{C}^\times \rightarrow \mathbb{R}^\times, z \mapsto |z|$? From $\mathbb{Z}^2 \rightarrow \mathbb{Z}, (m, n) \mapsto m - n$?

3 §1.4: from tables to presentations

A multiplication table for a group of size k has k^2 entries. For $\mathbb{Z}/3\mathbb{Z}$:

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

Fine for $k = 3$; hopeless for S_{10} ($k = 3,628,800$) and impossible for $\mathbb{Z}$.

4 Generators are not enough

F_2 and $\mathbb{Z}^2$ both have two generators. *Abelian* means $gh = hg$ for all g, h ; $\mathbb{Z}^2$ is, F_2 is not ($ab \neq ba$ as reduced words). So a generating set alone does not pin down a group. Something must be said about which products coincide.

5 Equations imply equations

The example: from $gh = k$ and $h = pq$ deduce $gpq = k$ by substitution. More generally, any equation can be multiplied on both sides by anything, inverted, and combined with others. A presentation lists a few equations from which all the rest follow by such manipulations.

6 $\mathbb{Z}/n\mathbb{Z}$ in multiplicative notation

Renaming k as a^k turns addition mod n into multiplication of powers with exponents mod n . This is the same group, and the change of notation makes the presentation on the next page read naturally.

Interlude: I. Pages 15–17 step by step: cosets in D_3 , a product that fails, and the quotient $D_3/N \cong \mathbb{Z}/2\mathbb{Z}$

D_3 has six elements: the rotations $1, r, r^2$ (blue) and the reflections s, sr, sr^2 (red), with $r^3 = 1, s^2 = 1$ and $rs = sr^2$. Every product can be read off the coloured table on visual page V1. Products are read right factor first, as always.

1. The cosets of the rotations N . Multiply every element of N by s on the left: $s \cdot 1 = s, s \cdot r = sr, s \cdot r^2 = sr^2$. Multiply on the right instead: $1 \cdot s = s, r \cdot s = sr^2, r^2 \cdot s = sr$. Same set both ways.

$N = \{1, r, r^2\}$, the rotations

$sN = \{s, sr, sr^2\}$, the reflections



two cosets, each of size 3; $6 = 2 \cdot 3$

So $sN = Ns$: left and right cosets agree, which is what “normal” means in practice. Conjugation check: $rsr^{-1} = rsr^2 = sr^2 \cdot r^2 = sr$, a reflection sent to a reflection, and $sr s^{-1} = sr s = r^2$, a rotation sent to a rotation.

2. Multiplying cosets by multiplying representatives. $N \cdot N$: pick r and r^2 , product $1 \in N$. $sN \cdot sN$: pick s and sr , product $s sr = r \in N$. $N \cdot sN$: pick r^2 and s , product $r^2 s = sr \in sN$. Whatever representatives you pick, the answer lands in the same coset, so the two cosets form a group with two elements:

$\cdot$	N	sN
N	N	sN
sN	sN	N

which is the table of $\mathbb{Z}/2\mathbb{Z}$: rotation = 0, reflection = 1.

3. A subgroup that is not normal, and the product that goes wrong (Exercise 9). Take $H = \{1, s\}$. Left cosets: $H, rH = \{r, rs\} = \{r, sr^2\}, r^2H = \{r^2, r^2s\} = \{r^2, sr\}$. Right cosets: $H, Hr = \{r, sr\}, Hr^2 = \{r^2, sr^2\}$. The partitions differ: $rH \neq Hr$.

Now try to multiply the classes of s and r by representatives.

- Representatives s and r : product sr , which lies in the coset $\{sr, r^2\}$.
- Other representatives of the same classes, $1 \in sH = \{s, 1\}$ and $sr^2 \in rH$: product $1 \cdot sr^2 = sr^2$, which lies in the coset $\{sr^2, r\}$.

Two different cosets from the same two classes: the product is not well defined, so D_3/H is not a group. The computation only works when $g_1 n_2 g_1^{-1}$ stays in the subgroup, which is exactly the normality condition, and $rsr^{-1} = sr \notin H$ is the culprit here.

4. The first isomorphism theorem, in this example. The map $f : D_3 \rightarrow \mathbb{Z}/2\mathbb{Z}$, “flipped or not”, sends rotations to 0 and reflections to 1; its kernel is N . The theorem’s map $\phi([g]) = f(g)$ sends the coset N to 0 and the coset sN to 1, matching the two-element table above with the table of $\mathbb{Z}/2\mathbb{Z}$ entry for entry: $D_3/N \cong \mathbb{Z}/2\mathbb{Z}$.

Try it

Do the same for $\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$: the cosets of $4\mathbb{Z}$ are the four colours on visual page V4, adding representatives ($5 + 7 = 12, 1 + 3 = 4$) always lands in the coset of 0, and the four cosets form $\mathbb{Z}/4\mathbb{Z}$.

presentation

$$\mathbb{Z}/n\mathbb{Z} \cong \langle a \mid a^n = 1 \rangle.$$

So the generating set is a and the only defining equality is $a^n = 1$. Every entry in the multiplication table for $\mathbb{Z}/n\mathbb{Z}$ follows from this one equality. For instance, one entry in the multiplication table is $a^{n-3}a^4 = a$. But we can easily derive this using the defining equality $a^n = 1$ as follows:

$$a^{n-3}a^4 = a^{n+1} = a^n a^1 = 1 \cdot a = a.$$

A second important example of a group presentation is the following:

$$\mathbb{Z}^2 = \langle x, y \mid xy = yx \rangle.$$

Can you see why this is a presentation of $\mathbb{Z}^2$? We'll come back to this in a little bit.

Exercise 11. Convince yourself that $\mathbb{Z}^2$ indeed has the purported presentation.

Hopefully these examples give you an idea of what a group presentation is. The way we make things formal—and the way we really think about group presentations—is through the use of free groups. At first our definition of a presentation won't look exactly like the examples we just gave, but we'll eventually explain why it is really the same thing.

3 A *group presentation* is a pair (S, R) , where S is a set and R is a set of words in S . If we freely reduce the elements of R , then we can regard them as elements of the free group $F(S)$, and indeed it will be important to take this point of view. The presentation is finite when both sets S and R are finite.

4 We say that a group G has the presentation $\langle S \mid R \rangle$ if G is isomorphic to the quotient of $F(S)$ by the normal closure of R , the smallest normal subgroup of $F(S)$ containing R (equivalently, the subgroup of $F(S)$ generated by all conjugates of all elements of R). We write:

$$G \cong \langle S \mid R \rangle.$$

Here is how we think of this: the group G consists of reduced words in $S \cup S^{-1}$ but with the added caveat that words in R are the same as the empty word (this is where the normal closure comes in: if some element of R is supposed to represent the identity, then any conjugate of that element in $F(S)$ should also represent the identity, and any products of conjugates of such elements...). The elements of S are called *generators* for G and the elements of the normal closure of R are called *relators* for G , and the elements of R are called defining relators. If we have a presentation (as in our above discussion) with a relation like $ab = ba$ or $aba = bab$, then we turn this into a relator by moving everything to one side: $aba^{-1}b^{-1}$ or $abab^{-1}a^{-1}a^{-1}$.

5 How does this formal definition match up with what we said before? Let's look more closely at $\mathbb{Z}^2$. This group has the presentation

$$\mathbb{Z}^2 \cong \langle a, b \mid aba^{-1}b^{-1} \rangle.$$

(Notice that this is different from the presentation we gave a few minutes ago, but bear with us!) What are the elements of the group $\langle a, b \mid aba^{-1}b^{-1} \rangle$? Well, we

$$\mathbf{1} \quad \langle a \mid a^n = 1 \rangle$$

Read: “the group generated by a , subject only to the rule $a^n = 1$ ”. Everything else must follow from that rule. The worked example $a^{n-3}a^4 = a$ shows how: collect exponents, then use $a^n = 1$ to reduce. Also from the rule: $a^{-1} = a^{n-1}$, so inverses need not be listed separately.

$$\mathbf{2} \quad \langle x, y \mid xy = yx \rangle$$

Two generators, one rule: they commute. Using the rule you can move every x to the left past every y , so any word becomes $x^m y^n$, and different (m, n) give different elements. That is $\mathbb{Z}^2$. Exercise 11 asks you to convince yourself of this; the formal version is worked on page 19.

$$\mathbf{3} \quad \text{The formal definition, } (S, R)$$

S is the alphabet, R a set of words (the rules, in the form “this word = 1”). The group is

$$\begin{array}{ccc} F(S) & \xrightarrow{\text{send each letter to itself}} & G \cong \langle S \mid R \rangle \\ \text{kernel} \uparrow & & \\ \langle\langle R \rangle\rangle = \text{normal closure of } R & & \end{array}$$

i.e. take the free group and squash the normal closure of R to the identity. “Finite presentation” means finitely many generators and finitely many rules; most groups in this book are finitely presented.

$$\mathbf{4} \quad \text{Why the normal closure and not just the subgroup generated by } R?$$

Why?

If a word r is to equal 1, then so must grg^{-1} for every g (since $g \cdot 1 \cdot g^{-1} = 1$), and products and inverses of such elements too. The set of all consequences is therefore a normal subgroup containing R , and the smallest such is the normal closure $\langle\langle R \rangle\rangle$. Quotienting by a non-normal subgroup would not even give a group (page 16, **4**).

So the elements of $\langle S \mid R \rangle$ are reduced words, with two words equal exactly when one can be turned into the other by inserting or deleting conjugates of relators.

$$\mathbf{5} \quad \text{Relators versus relations}$$

A *relation* is an equation $u = v$; the corresponding *relator* is uv^{-1} (so that $uv^{-1} = 1$). From $ab = ba$: $ab(ba)^{-1} = aba^{-1}b^{-1}$.

$$\mathbf{6} \quad \text{A slip in the printed text}$$

Watch out

From $aba = bab$ the relator is $aba(bab)^{-1} = abab^{-1}a^{-1}b^{-1}$. The page prints the last letter as a^{-1} ; it should be b^{-1} . (Apply socks-and-shoes to bab : its inverse is $b^{-1}a^{-1}b^{-1}$.) This relation, $aba = bab$, is the braid relation of Office Hour 18.

$$\mathbf{7} \quad \mathbb{Z}^2 \cong \langle a, b \mid aba^{-1}b^{-1} \rangle$$

Same group as **2**, now written with a relator instead of a relation. The next page proves the two descriptions agree.

start with the free group on a and b , which consists of all freely reduced words in a and b . Then we need to take the quotient by the normal closure of $aba^{-1}b^{-1}$.

- 1 What is the quotient as a set? Well, in the quotient we can change a word w into an equivalent word by inserting $aba^{-1}b^{-1}$ anywhere in w , or removing it from anywhere in w .

Why is that the same as our first presentation of $\mathbb{Z}^2$ as $\langle x, y \mid xy = yx \rangle$? Well, we claim that two elements of the free group on $\{a, b\}$ are equivalent in the quotient (by the normal quotient by $aba^{-1}b^{-1}$) if and only if they differ by a sequence of swaps, where we swap ab for ba or vice versa. Let's convince ourselves of this.

- 2 Start with a word $wba w'$, where w and w' are both words in a and b . We said we can insert $aba^{-1}b^{-1}$ anywhere we want; so in the quotient,

$$wba w' = w(aba^{-1}b^{-1})ba w' = wab(a^{-1}b^{-1}ba)w' = wab w'.$$

- 3 By applying swaps like this we see that every element of the quotient is equivalent to $a^m b^n$ for some m and n . But this gives us an obvious isomorphism between our quotient and $\mathbb{Z}^2$; the isomorphism is given by $a^m b^n \leftrightarrow (m, n)$.

We can also see from the above discussion how our new presentation of $\mathbb{Z}^2$ (with generators a and b) matches up with our old one (with generators x and y). Both presentations are really saying the same thing. So which one you use just depends on your preference. When we have an equality between generators like $xy = yx$, then the equality is called a *relation*. And when we have an equality like $aba^{-1}b^{-1} = 1$ we say that $aba^{-1}b^{-1}$ is a *relator*, as in our formal definition. We can always change a relation to a relator by moving all of the generators to one side (as in our $\mathbb{Z}^2$ example), and the net effect is the same. In this book we will tend to use relators more than relations, but both will come up.

- 4 Every group can be given by a group presentation. In particular:

Every group is a quotient of a free group.

This is one reason why free groups are so important. In many ways, geometric group theory grew out of the study of combinatorial group theory, a subject largely concerned with studying groups via their presentations.

- 5 **Exercise 12.** Show that every group has a presentation. *Hint: Your presentation does not have to be efficient.*
- 6 **Exercise 13.** Find (nice) group presentations for F_n , D_n , and S_n .

Armed with the notion of a group presentation, you can start to list lots of different groups. Which presentations represent isomorphic groups? It turns out that this is an unsolvable problem in general; there is no algorithm that will take two presentations and decide in finite time if they present the same group. It is good to keep this in mind: group presentations are very helpful, but they do have limitations.

That's it for the first office hour—we introduced a number of examples of groups that will play key roles in the rest of the book. We want to understand these groups from as many points of view as possible. At first glance you might wonder what there is to learn about a free abelian group or a free group. The answer: more than you might imagine!

1 Inserting or deleting the relator anywhere

Why “anywhere”? Inserting r between w and w' gives $wrw' = (wrw^{-1})ww'$, and wrw^{-1} is a conjugate of r , hence in the normal closure, hence trivial in the quotient. So insertion at any position is allowed, and deletion is its inverse.

2 The swap computation, annotated

$$wba w' = w(aba^{-1}b^{-1})ba w' = wab \underbrace{(a^{-1}b^{-1}ba)}_{\text{reduces to 1}} w' = wab w'.$$

Step 1 inserts the relator; step 2 regroupes; step 3 cancels $b^{-1}b$ then $a^{-1}a$. Read backwards, ab can be turned into ba . So the single relator $aba^{-1}b^{-1}$ lets a and b pass through each other, which is the rule $ab = ba$.

3 Every element is $a^m b^n$

Repeatedly swap adjacent ba pairs to ab (bubble sort again: move all $a^{\pm 1}$ to the left), then collect exponents. Distinct (m, n) give distinct elements because the exponent-sum map $F_2 \rightarrow \mathbb{Z}^2$ of page 15, item 7, sends $a^m b^n \mapsto (m, n)$ and kills the relator, so it descends to the quotient and separates them. With Theorem 1.1: the quotient is $F_2/[F_2, F_2] \cong \mathbb{Z}^2$, item 7 on page 17.

4 “Every group is a quotient of a free group”

Why?

Proof (this is Exercise 12). Let G be any group and take $S = G$, the whole group as alphabet. Define $F(G) \rightarrow G$ by sending each letter to the element it names and each word to the product; this is a homomorphism by construction and it is onto (the one-letter word g maps to g). Let R be its kernel. By Theorem 1.1, $G \cong F(G)/R$, so $G \cong \langle G \mid R \rangle$. Wasteful, but a presentation.

Combinatorial group theory, the older subject mentioned here, studies groups through presentations; geometric group theory adds pictures.

5 Exercise 12

See 4. For a small finite group one can take R to be the whole multiplication table, written as relators $gh(gh)^{-1}$... note that for $S = G$ the letter “ gh ” and the word “ g followed by h ” are different things.

6 Exercise 13: hints

$F_n = \langle a_1, \dots, a_n \mid \rangle$ (no relators). $D_n = \langle s, t \mid s^2, t^2, (st)^n \rangle$ using the two reflections of page 6, or $\langle r, s \mid r^n, s^2, srs^{-1}r \rangle$ using a rotation and a reflection. $S_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i^2, (\sigma_i \sigma_{i+1})^3, (\sigma_i \sigma_j)^2 \text{ for } |i-j| \geq 2 \rangle$ with $\sigma_i = (i \ i+1)$; the middle relator encodes $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$, the braid relation from page 18. Proving these are complete presentations is harder than writing them down.

7 Unsolvable in general

Not “hard”: provably impossible. There is no algorithm that decides whether two finite presentations give isomorphic groups (Adian and Rabin, 1950s). Even the *word problem*, deciding whether a given word equals 1 in $\langle S \mid R \rangle$, is unsolvable for some finite presentations (Novikov, Boone).

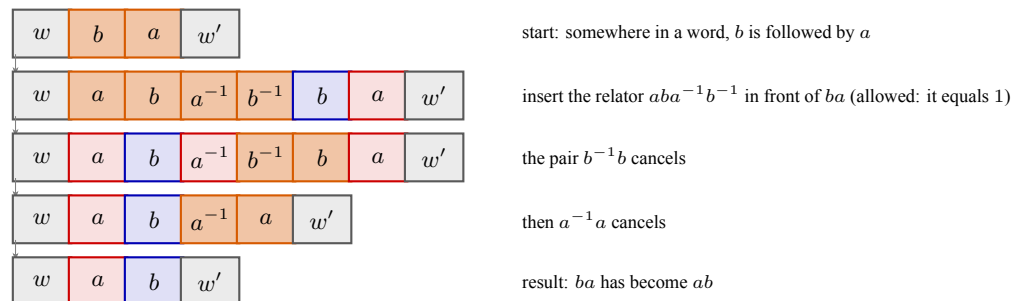
Looking ahead

Office Hour 9 shows that for *hyperbolic* groups the word problem is solvable, and quickly. Much of geometric group theory is about which geometric properties of a group make such problems tractable.

Interlude: J. Pages 18–19 step by step: what a relator lets you do, and why $\langle a, b \mid aba^{-1}b^{-1} \rangle$ is $\mathbb{Z}^2$

In $\langle S \mid R \rangle$ the words in R are declared equal to 1, and so are all their conjugates and all products of those. Concretely: you may insert or delete a relator anywhere in a word without changing the element. Boxes are letters, as in Interlude G; highlighted boxes are the ones about to be inserted or cancelled.

1. One relator, one swap.



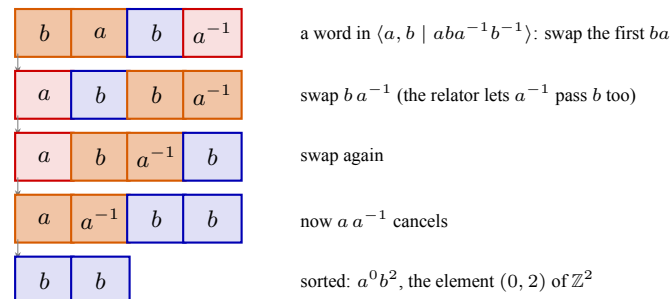
Why?

Why “anywhere” is allowed: inserting r between w and w' gives $wrw' = (wrw^{-1})ww'$, and wrw^{-1} is a conjugate of r , hence in the normal closure, hence trivial in the quotient. This is exactly why the definition uses the normal closure rather than just the subgroup generated by R .

Watch out

The relator does not say “delete b and a ”; it says they may change places. Every letter survives, only the order changes, which is why the exponent sums of a and of b never change. That is the map $F_2 \rightarrow \mathbb{Z}^2$ of page 15 in action.

2. Sorting any word into $a^m b^n$ (Exercise 11). Repeating the swap moves every $a^{\pm 1}$ to the left of every $b^{\pm 1}$; then ordinary cancellation collects the exponents.



Every element is therefore some $a^m b^n$, and $a^m b^n \leftrightarrow (m, n)$ is the isomorphism with $\mathbb{Z}^2$: multiplying $a^m b^n$ by $a^{m'} b^{n'}$ and sorting gives $a^{m+m'} b^{n+n'}$, which is vector addition. Different pairs (m, n) give different elements because the exponent-sum map already separates them.

3. $\mathbb{Z}/n\mathbb{Z} = \langle a \mid a^n \rangle$, in the same style. With $n = 5$: $a^2 \cdot a^4 = a^6$; delete the relator a^5 from $a^6 = a^5 a$ and get a . On the pentagon of Interlude E this is “2 clicks then 4 clicks is 1 click”. Every entry of the multiplication table follows from the single rule $a^5 = 1$.

Try it

Exercise 12 in miniature: present $\mathbb{Z}/3\mathbb{Z} = \{0, 1, 2\}$ with the whole group as alphabet, $S = \{x_0, x_1, x_2\}$, and one relator for each table entry, $x_1 x_1 x_2^{-1}$, $x_1 x_2 x_0^{-1}$, and so on, plus x_0 itself. Wasteful, but $\langle S \mid R \rangle$ is $\mathbb{Z}/3\mathbb{Z}$: every letter can be rewritten as a power of x_1 , and x_1^3 reduces to $x_0 = 1$.

- 1 Hopefully we have convinced you that the concept of a group is deeply intertwined with the idea of symmetries: groups lead to symmetries and symmetries lead to groups. In order to study infinite groups, we will want to take advantage of this and think of groups geometrically. A key step on this path is to show that every group really is the group of symmetries of some geometric object. That's the goal of the next office hour—see you there!
- 2

1 “Groups lead to symmetries and symmetries lead to groups”

Here is the table the Big-picture box on page 3 asked you to build, filled in from the chapter:

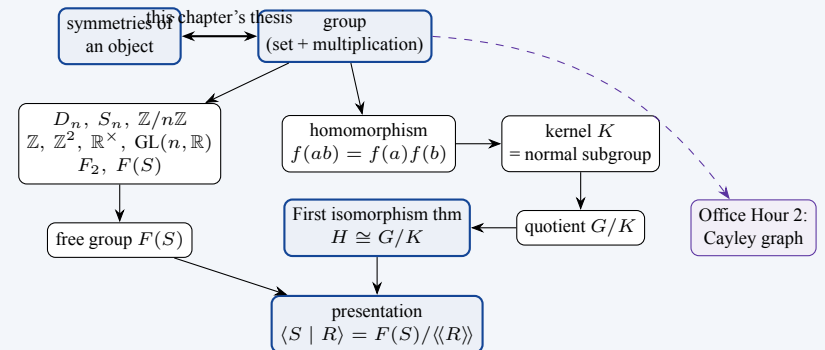
group	multiplication	symmetries of...
S_4	composition	rotations of the cube
D_n	composition	regular n -gon
S_n	composition	n loose points
$\mathbb{Z}/n\mathbb{Z}$	add mod n	n -gon with arrowed edges
$\mathbb{Z}$	add	line with marked points and arrows
$\mathbb{Z}^2$	add	grid with coloured arrows
$\mathbb{R}^\times, \mathbb{C}^\times$	multiply	(scalings of a line / plane)
$GL(n, \mathbb{R})$	matrix product	linear maps of $\mathbb{R}^n$
F_2	concatenate, reduce	the 4-valent tree (Interlude B)
$\langle S \mid R \rangle$	words mod relators	its Cayley graph (Office Hour 2)

2 What Office Hour 2 will do

Given any group G with generating set S , draw one dot per element and an arrow labelled s from g to gs . The result is the Cayley graph. Theorem 2.2 says G is exactly the group of label-preserving symmetries of this graph. The arrowed polygon, the arrowed line, the coloured grid and the tree are the four Cayley graphs you have already seen.

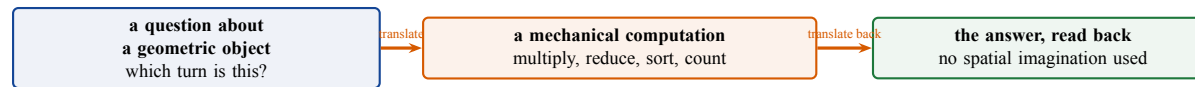
Big picture

Chapter map. Everything in the chapter fits on one diagram:



Interlude: P. Algebra making geometry computable: one pattern, six instances

Office Hour 1 runs in one direction throughout: take a question you would otherwise have to answer by turning an object in your hands, and convert it into a calculation that a careful reader can finish on paper without picturing anything. Every section of the chapter supplies one instance of the same pattern.



1. Composing two turns of a cube

The question. A quarter turn about the vertical axis, then a quarter turn about the left–right axis: which single rotation is that?

The translation. Each turn is a permutation of the four long diagonals, and each is a four-cycle.

The computation. Multiply the two cycles.

The answer. A three-cycle, so the result is a third of a turn about a long diagonal, the one through the corner the cycle fixes. Interlude K draws it, but the algebra found it.

2. Composing two reflections of a pentagon

The question. Reflect in one axis, then in another that makes an angle of 36° with it. What has happened to the pentagon?

The computation. The product is the word st , and st is one click.

The answer. A rotation by 72° . The general rule, that reflecting in two lines meeting at θ rotates by 2θ , is read off the word rather than from a diagram.

3. Telling the time

The question. What time is it four hours after nine o'clock?

The computation. $9 + 4 = 13$ in $\mathbb{Z}$, then subtract 12.

The answer. One o'clock. Everyone does this without thinking, which is the point: $\mathbb{Z}/n\mathbb{Z}$ is the arithmetic you already perform whenever a quantity wraps around.

4. Does a walk return home?

The question. Walk on the four-valent tree following the instructions $a b a^{-1} b^{-1} b a b^{-1} a^{-1}$. Do you finish where you started?

The computation. Reduce the word by deleting adjacent inverse pairs.

The answer. Yes exactly when the reduced word is empty, and for a tree that is the only way home. A question about a path in an infinite picture becomes a finite cancellation.

5. The same walk on the grid

The question. The same instructions, now on the coloured grid.

The computation. Use the relator $aba^{-1}b^{-1}$ to push every a past every b , then collect exponents into $a^m b^n$.

The answer. Home exactly when $m = n = 0$. Notice that the word above returns home on the grid but not necessarily on the tree, which is the difference between $\mathbb{Z}^2$ and F_2 in one sentence.

6. Even or odd?

The question. Can a given rearrangement of six objects be achieved by an even number of swaps?

The computation. Apply the sign homomorphism: a k -cycle costs $k - 1$ swaps, so $(1\ 6\ 2\ 4)(3\ 5)$ costs $3 + 1 = 4$.

The answer. Even. A question that sounds like it needs a search is settled by one addition, because a homomorphism to a two-element group cannot be fooled.

Big picture

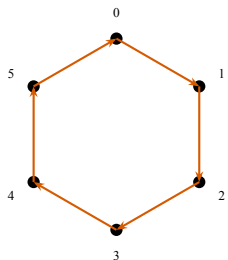
And the axioms alone already pay. Before any example is chosen, the three axioms give: the identity is unique; inverses are unique; $(gh)^{-1} = h^{-1}g^{-1}$; $gx = k$ has exactly one solution, namely $g^{-1}k$; and $gx = gy$ forces $x = y$. Each is proved in two lines and each holds in every group in this chapter and every group you will meet later.

Looking ahead

Office Hour 2 runs the arrow the other way. There one assumes a group acts on a space in some prescribed manner and deduces an algebraic fact about the group, with no geometry left in the conclusion. The two directions together are what the subject is made of.

Interlude: V5. A gallery of Cayley graphs: what presentations look like

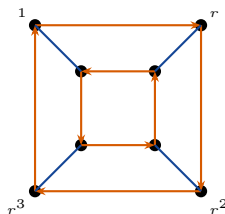
One dot per group element; from each dot g , one arrow per generator x pointing to gx (an arrow is drawn two-way when $x = x^{-1}$). Reading the letters along any path spells a word; a closed loop spells a word equal to 1, i.e. a relator.



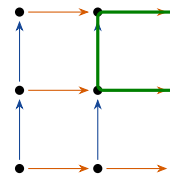
$\mathbb{Z}/6\mathbb{Z} = \langle a \mid a^6 \rangle$
the relator is the loop around



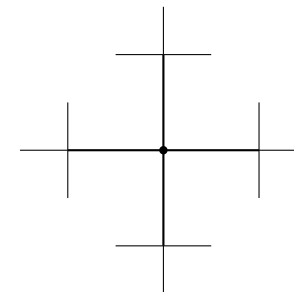
$\mathbb{Z} = \langle a \mid \rangle = F_1$
no relator, no loop



$D_4 = \langle r, s \mid r^4, s^2, (sr)^2 \rangle$
orange = r , blue = s (two-way)



$\mathbb{Z}^2 = \langle a, b \mid aba^{-1}b^{-1} \rangle$
every square is the relator



$F_2 = \langle a, b \mid \rangle$
no relators, no loops: a tree

Why?

Why does the group act as symmetries of its own graph (Theorem 2.2)? Multiplying every element on the *left* by a fixed h sends $g \mapsto hg$ and $gx \mapsto hgx$, so it carries each arrow $g \rightarrow gx$ to the arrow $hg \rightarrow hgx$ with the same label. Every element of the group is therefore a label-preserving symmetry of the picture, and one checks that these are all of them. That is the theorem announced on page 5, one office hour early.

Try it

In the D_4 graph, start at 1 and read the word $s r s r$ (blue, orange, blue, orange). You should return to 1: this is the relator $(sr)^2$, which encodes $srs = r^{-1}$, “a reflection reverses the direction of rotation”. Then check that $r s r s$ also closes up, and that $r r s$ does not.

Looking ahead

The shapes are the point. $\mathbb{Z}$ looks like a line, $\mathbb{Z}^2$ like a plane, F_2 like a tree, and finite groups like bounded blobs. Office Hour 7 defines when two such pictures “look the same from far away” and shows this large-scale shape is a property of the group, not of the generators chosen.

Closing pages: all thirteen exercises at a glance, and a glossary

1. **(p. 8)** Adjacent transpositions generate S_n . Draw the permutation with straight strands, separate crossing heights, read off the crossings. Equivalent to bubble sort.
2. **(p. 8)** S_n as symmetries of a shape: the regular simplex with n corners (tetrahedron for $n = 4$, including reflections). Higher cubes give other groups.
3. **(p. 8)** $\gcd(m, n) = 1 \Rightarrow \{m\}$ generates $\mathbb{Z}/n\mathbb{Z}$: Bézout gives $um \equiv 1$.
4. **(p. 12)** Matrix group over a finite field: $\mathrm{GL}(2, \mathbb{Z}/p\mathbb{Z})$; for $p = 2$ it has 6 elements and is S_3 in disguise.
5. **(p. 12)** Uniqueness of the reduced word: induction on length via the two-cases argument of Exercise 6.
6. **(p. 13)** Associativity of F_2 : overlapping and non-overlapping cancellations both lead to the same word (diamond lemma).
7. **(p. 14)** More injective homomorphisms: $D_n \rightarrow S_n$ (act on vertices), $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/mk\mathbb{Z}$ ($1 \mapsto k$), $F_2 \rightarrow F_3$, $F_2 \rightarrow F_2$ ($a \mapsto a^2$, $b \mapsto b$), $\mathbb{Z}^2 \rightarrow \mathrm{GL}(2, \mathbb{R})$ via $(m, n) \mapsto \mathrm{diag}(2^m, 3^n)$.
8. **(p. 15)** More non-injective homomorphisms: $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ (remainder), $\mathbb{Z}^2 \rightarrow \mathbb{Z}$ (sum of coordinates), $F_2 \rightarrow \mathbb{Z}/2\mathbb{Z}$ (parity of word length), $S_n \rightarrow S_{n+1} \rightarrow \mathbb{Z}/2\mathbb{Z}$, $D_n \rightarrow D_m$ when $m \mid n$.
9. **(p. 16)** G/N well defined $\iff N$ normal: the one-line computation $n_1 g_1 n_2 g_2 = n_1 (g_1 n_2 g_1^{-1}) g_1 g_2$, both directions.
10. **(p. 17)** Your own isomorphisms: $\mathbb{C}^\times / S^1 \cong \mathbb{R}^+$ (absolute value), $\mathbb{Z}^2 / \{(m, m)\} \cong \mathbb{Z}$ (difference), $\mathrm{GL}(n, \mathbb{R}) / \mathrm{GL}^+(n, \mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z} / \{0, 3\} \cong \mathbb{Z}/3\mathbb{Z}$.
11. **(p. 18)** $\langle x, y \mid xy = yx \rangle \cong \mathbb{Z}^2$: sort each word to $x^m y^n$; the exponent-sum map shows distinct pairs stay distinct.
12. **(p. 19)** Every group has a presentation: $F(G) \rightarrow G$ is onto; apply Theorem 1.1.
13. **(p. 19)** $F_n = \langle a_1, \dots, a_n \mid \rangle$; $D_n = \langle s, t \mid s^2, t^2, (st)^n \rangle$; S_n with generators $\sigma_i = (i \ i+1)$ and relators σ_i^2 , $(\sigma_i \sigma_{i+1})^3$, $(\sigma_i \sigma_j)^2$ for $|i - j| \geq 2$.

Glossary (page of first appearance)

symmetry (3)	a rigid motion returning an object to its place
group (5)	set with an associative multiplication, identity, inverses
abelian (17)	$gh = hg$ for all g, h
generate (6)	every element is a product of the generators and inverses
D_n (6)	symmetries of the regular n -gon; $2n$ elements
S_n (6)	all permutations of n things; $n!$ elements
cycle, transposition (7)	$(i_1 \cdots i_k)$; a 2-cycle
$\mathbb{Z}/n\mathbb{Z}$ (8)	remainders mod n under addition
subgroup (9)	subset closed under product and inverse, containing 1
finitely generated (11)	has a finite generating set
$\mathrm{GL}(n, \mathbb{R})$, $\mathrm{SL}(n, \mathbb{R})$ (11)	invertible matrices; those of determinant 1
word, reduced word (12)	string over $S \cup S^{-1}$; no adjacent xx^{-1}
free group $F(S)$, rank (12–13)	reduced words, concatenate-then-reduce; $ S $
homomorphism (13)	$f(ab) = f(a)f(b)$
isomorphism (14)	bijective homomorphism; “same group”
injective / surjective (14)	one-to-one / onto
kernel (15)	elements sent to 1
normal subgroup (15)	$gng^{-1} \in N$ for all $g \in G$, $n \in N$
conjugate (15)	gng^{-1}
commutator subgroup (16)	kernel of $F_2 \rightarrow \mathbb{Z}^2$; generated by $xyx^{-1}y^{-1}$
quotient G/N (16)	classes $[g]$, $[g_1][g_2] = [g_1g_2]$
First Isomorphism Thm (17)	$f : G \twoheadrightarrow H$ with kernel K gives $H \cong G/K$
presentation $\langle S \mid R \rangle$ (18)	$F(S)$ modulo the normal closure of R
relation / relator (19)	$u = v$ / the word uv^{-1}
normal closure (18)	smallest normal subgroup containing R
word problem, isomorphism problem (19)	decide $w = 1$? decide $\langle S \mid R \rangle \cong \langle S' \mid R' \rangle$? unsolvable in general
Cayley graph (5, 20)	one vertex per element, one arrow per generator; Office Hour 2